

Active Transportation & the Built Environment

Accessibility, Design, and Planning

Fall 2026

Maryam Hosseini

City & Regional Planning

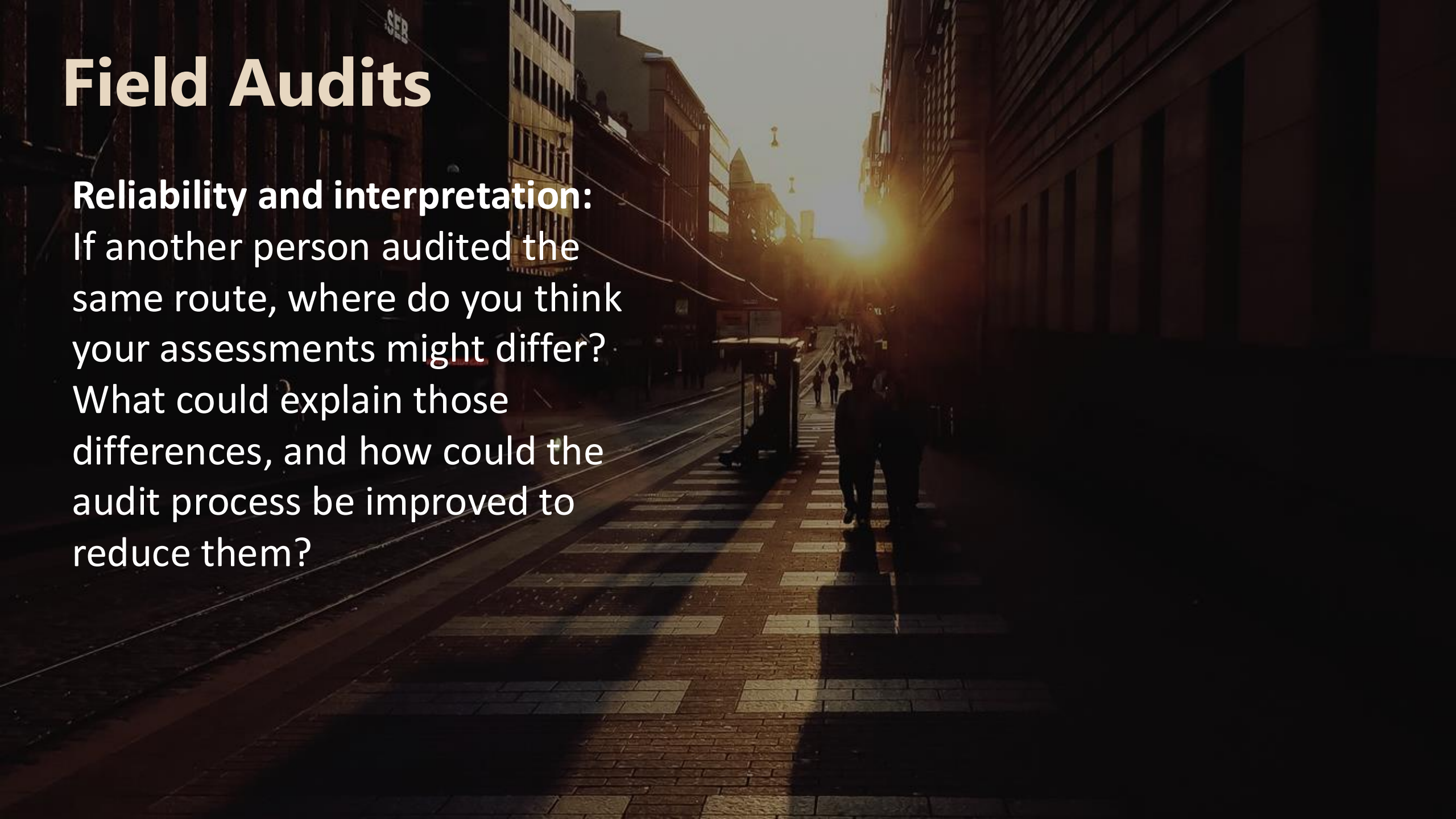
Field Audits

Your experience:
What was the most challenging part of conducting the audit? Were there conditions that were difficult to observe, classify, or assess using the audit tool?

What the tool captures:
Based on your field observations, what aspects of pedestrian accessibility did the audit tool capture well, and what important conditions did it overlook? Give specific examples.

Field Audits

Reliability and interpretation:
If another person audited the same route, where do you think your assessments might differ? What could explain those differences, and how could the audit process be improved to reduce them?



Limit of Detection (LOD)

- What is the ground truth?
- What is considered accurate?
- What is the precision level we need to answer a specific question?
- Tools have LOD, we have limit of measurement/quantification (LOQ)




STOOL Thousands
ENTRANCE
VALTIR
YODOCK
VALTIR


Safian Rudolph
JEWELERS
ESTD 1952
Buy • Sell • Repair
Diamonds • Jewelry
• Watches •
Gold • Platinum • Silver

“Wherever driving is the only practical means of getting around, as in much of urban America, the decision to drive may be **no choice at all**, but a **surrender to circumstances**”

- **Peter Norton**



Any comprehensive analysis
of pedestrian infrastructure
must include a threefold understanding of



Where

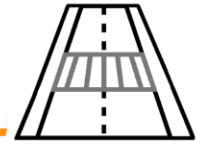
sidewalks are



How

they are connected

What



their condition is

Comprehensive Analysis

City Scale

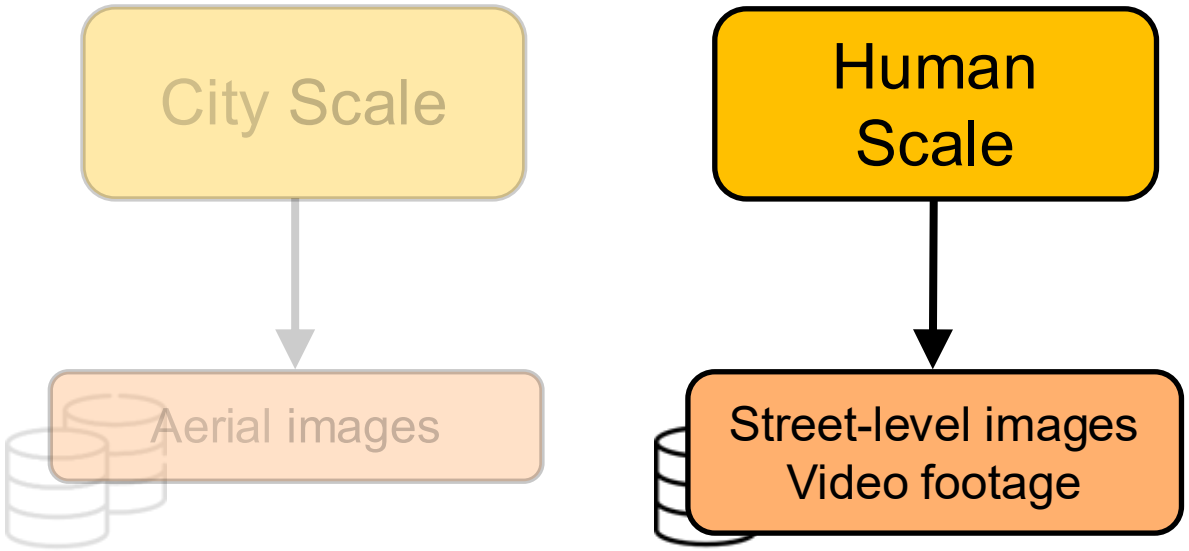
Aerial images

Human Scale

Street-level images
Video footage



Comprehensive Analysis







Spatial Structure of Sidewalks
City Scale

Mapping the Walk:

A Scalable Computer Vision Approach for Generating Sidewalk Network Datasets from Aerial Imagery

Open source: https://github.com/VIDA-NYU/tile2net

Hosseini, Maryam, Andres Sevtsuk, Fabio Miranda, Roberto M. Cesar Jr, and Claudio T. Silva.: Mapping the walk: A scalable computer vision approach for generating sidewalk network datasets from aerial imagery. Computers, Environment and Urban Systems, 2023

Tile2Net

- End-to-end open-source tool for creating pedestrian networks from orthorectified aerial imagery
- Scalable, easy to implement, and low-cost
- Despite the complexity of the code and process behind it, our user-friendly design makes the network generation process accessible to users, who can run it easily from the command line using

just one line of code!

Tile2Net on GitHub



Massachusetts Institute of Technology

Education Research Innovation Admissions + Aid Campus Life News Alumni About MIT

MIT News

ON CAMPUS AND AROUND THE WORLD

SUBSCRIBE BROWSE

Where the sidewalk ends

Most cities don't map their own pedestrian networks. Now, researchers have built the first open-source tool to let planners do just that.

Peter Dizikes | MIT News Office
March 15, 2023



MIT researchers h
source tool that u
image-recognitio
sidewalks and cr
Boston with an in
detected sidewall

Image: Inset courtes
via Google Earth

It's easier than ever to view maps of any place you'd like to go — by
another matter. Most cities and towns in the U.S. do not have sidev
pedestrians are usually left to fend for themselves: Can you walk fr
restaurants on the other side of the highway? Is there a shortcut fr
sports arena? And how do you get to that bus stop, anyway?

PETITION UPDATE

Select Board Approval- Mapping by MIT Tile2Net -Now to MassDOT

Kristin Alexander
United States



JUL 11, 2023 —
Falmouth Select Board has sent a letter of approval (see below) to MassDOT. In addition the Woods Hole Community Association and Senator Moran have sent letters.

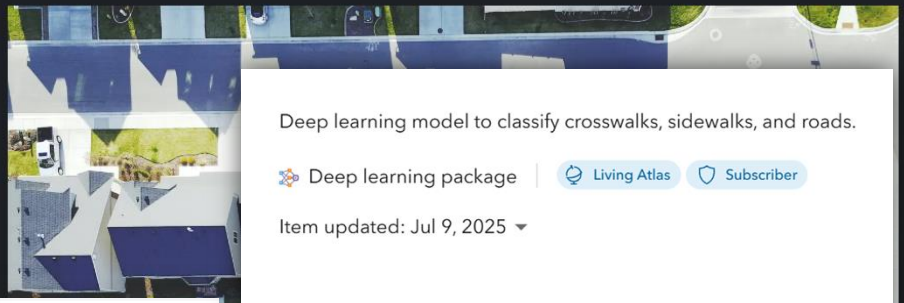
STREETSBLOG USA

USA NYC MASS LA CHI SF CAL

MAPS

New Tool Maps Sidewalks From the Sky — And Encourages Cities to Fill The Gaps

6:42 AM EDT on March 27, 2023



Deep learning model to classify crosswalks, sidewalks, and roads.

Deep learning package | Living Atlas | Subscriber

Item updated: Jul 9, 2025

View metadata

Owner

Esri Analytics Team
esri_analytics

Details

Number of downloads
70,416

Size
787.203 MB

Item ID
c0d520baa30d4b47ab362322...



Sharing

Everyone (public)

Acknowledgments

Hosseini, M., Sevtsuk, A., Miranda, F., Cesar, R. M., & Silva, C. T. (2023). Mapping the walk: A scalable computer vision approach for generating sidewalk network datasets from aerial imagery. Computers Environment and Urban Systems, 101, 101950.

Planetizen

Mapping Sidewalks for Improved Connectivity

A new tool uses aerial image recognition to map a city's sidewalks and crosswalks. Its developers hope it will aid in creating a more comprehensive understanding of pedestrian networks and where improvements are needed.

1 Minute Read
March 22, 2023, 12:00 PM PDT
By Diana Ionescu @aworkoffiction

Orthorectified Tiles

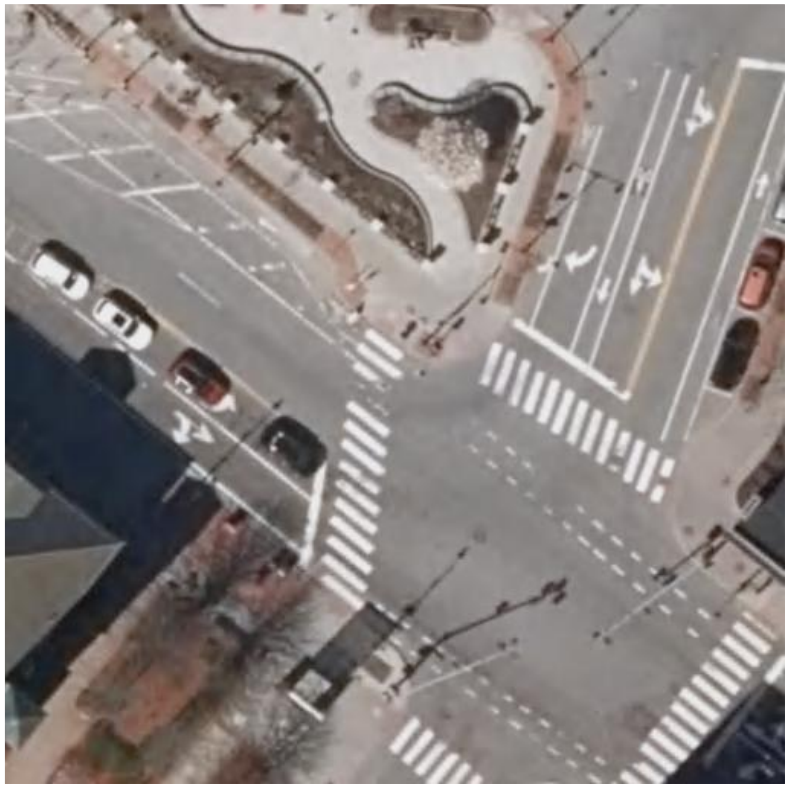
- Raw aerial images inherently contain distortions caused by:
  - Sensor orientation
  - Systematic sensor and platform-related geometry errors
  - Terrain relief
  - Curvature of the earth
- Orthorectification:
  - Removes inherent distortion
  - Creates accurately georeferenced images with uniform scale and consistent geometry
- Orthorectified imagery come in tile format.
- Tile system makes it possible to convert between positional coordinates of tiles and geographical coordinates.



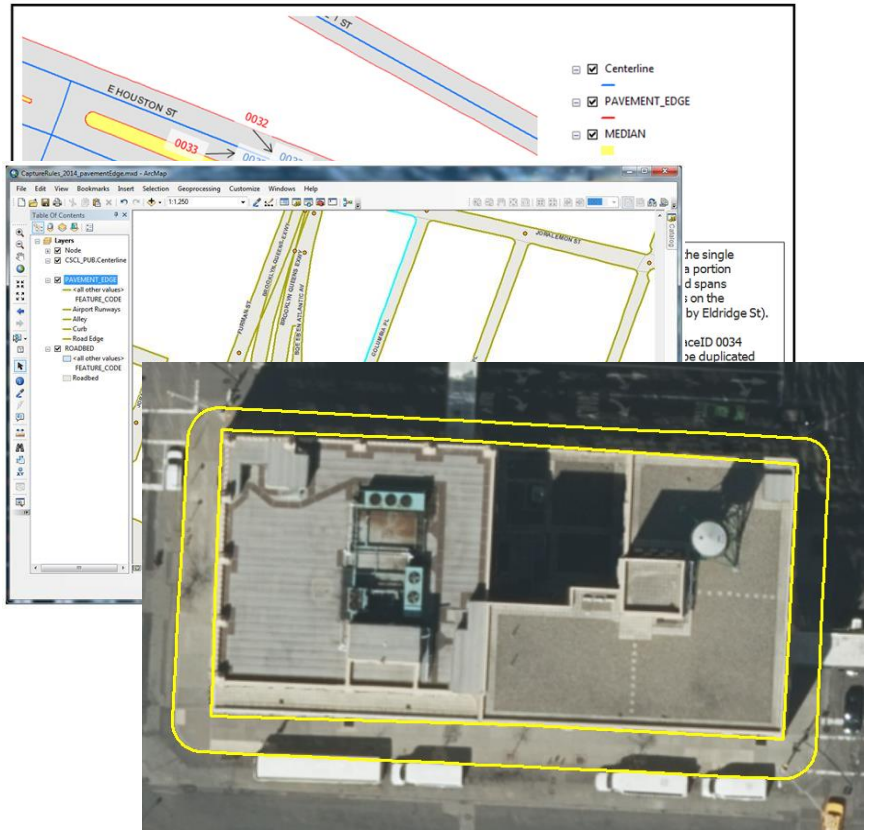
X, Y, Z system
X,Y: position of the tile on the grid
Z: zoom level
Higher Z → smaller area with more details

Semantic Segmentation

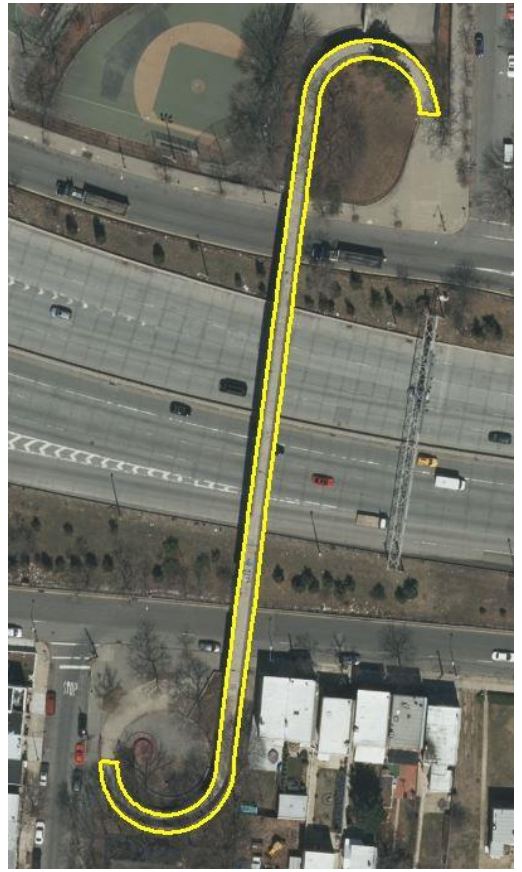
Input Image



Planimetric GIS Data

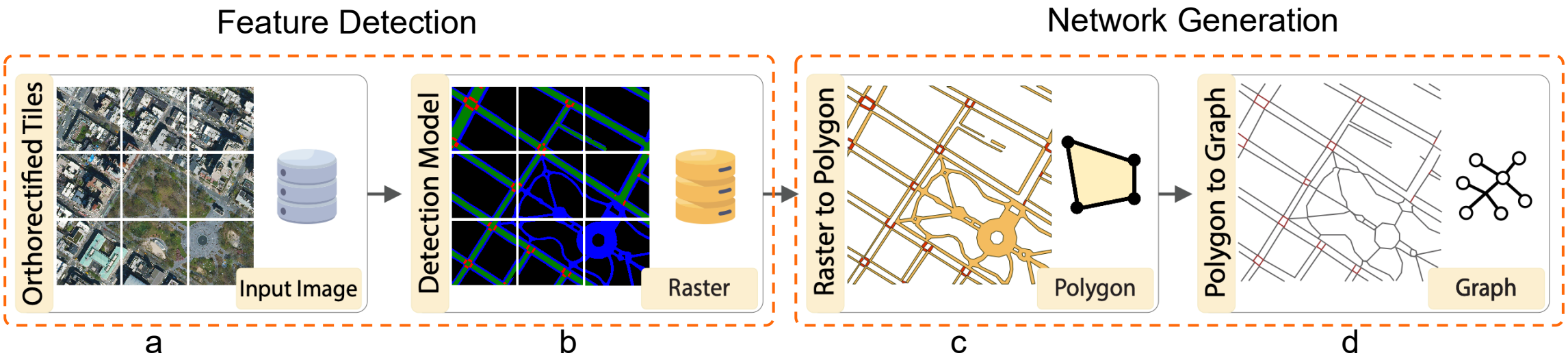






Pictures: NYC-Planimetric https://github.com/CityOfNewYork/nyc-planimetrics

Tile2Net: Pipeline



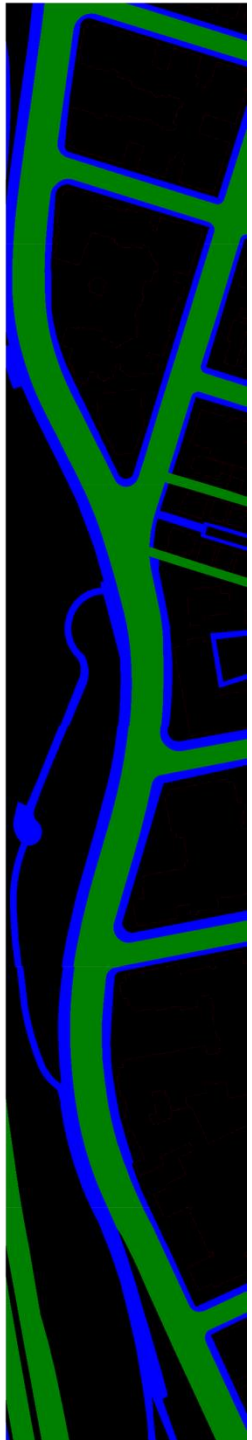
a) The unlabeled orthorectified tiles are passed through the semantic segmentation model for prediction

b) The model predicted the sidewalk (blue), crosswalks (red), and roads (green) for the input tiles

c) The prediction results (raster format) are converted into georeferenced polygons

d) The line representation of the pedestrian network generated from polygons





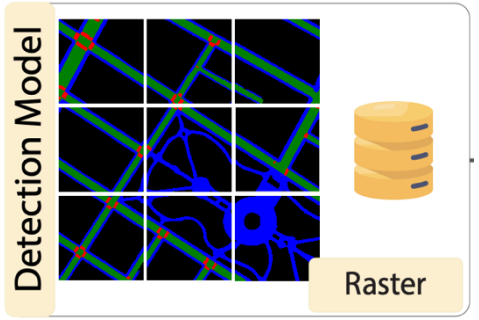
Sidewalk Detection Challenges

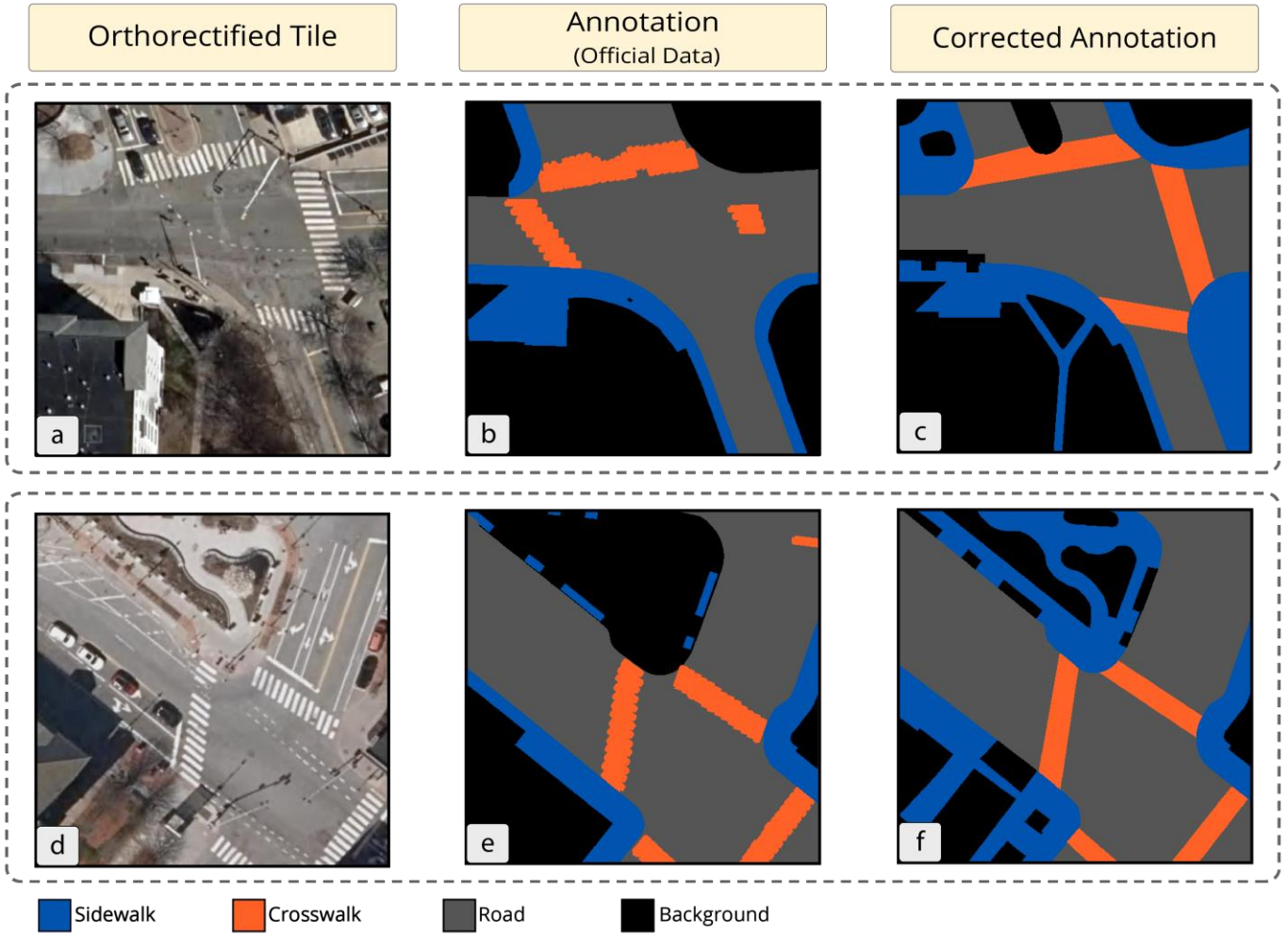
- Very fine and narrow area, less than 20% of images
- Often obstructed by bridges, vegetations, or shadow
- Different paving materials pose more challenges for the detection

Model: Hierarchical Multiscale Attention model

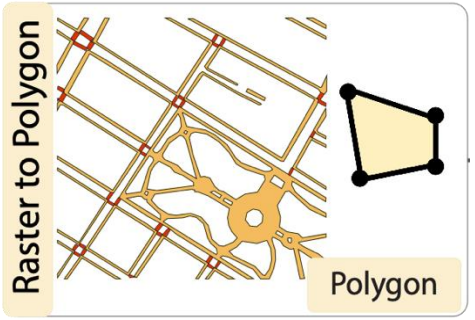
Data: High-resolution orthorectified images & Planimetric data

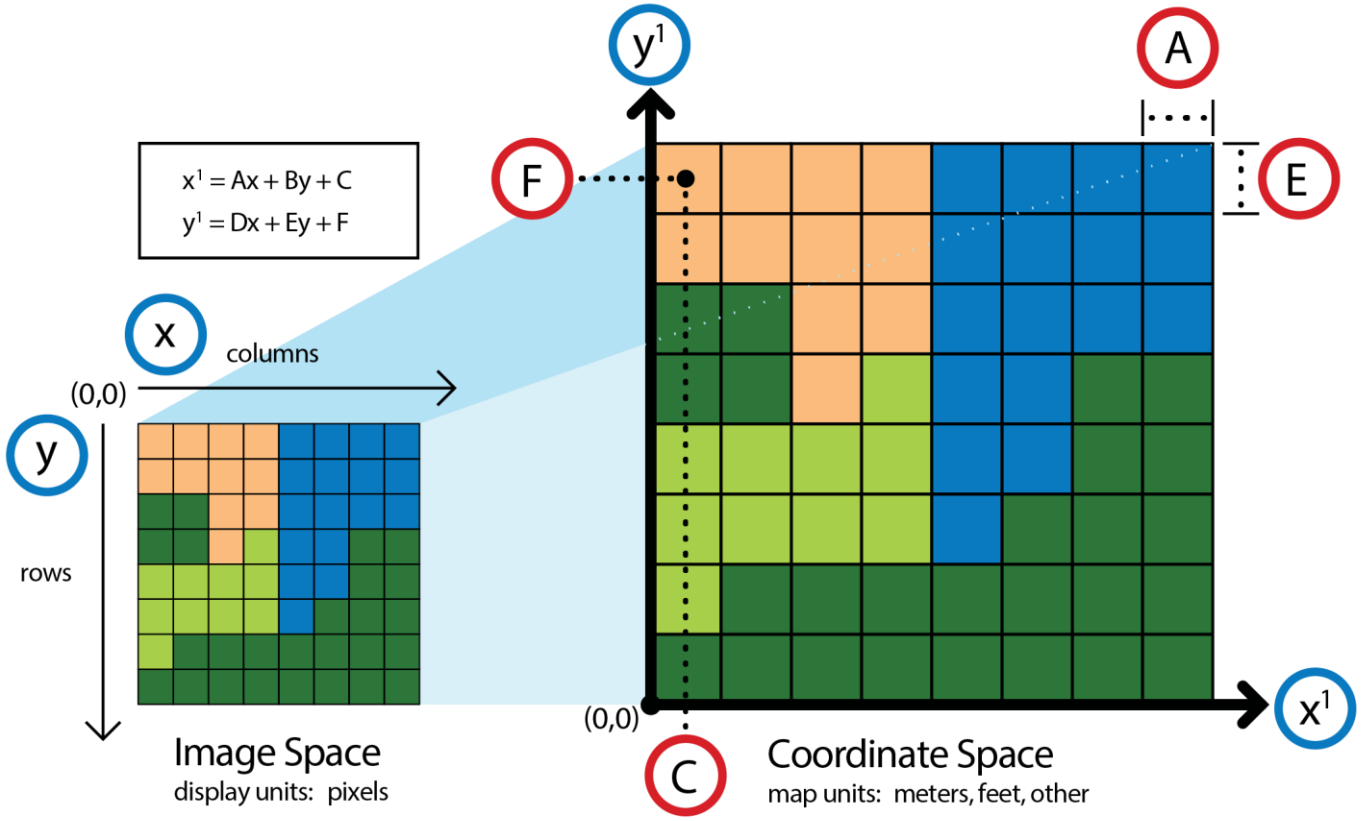
Problems in Official Data





Raster to Geo-referenced Polygon

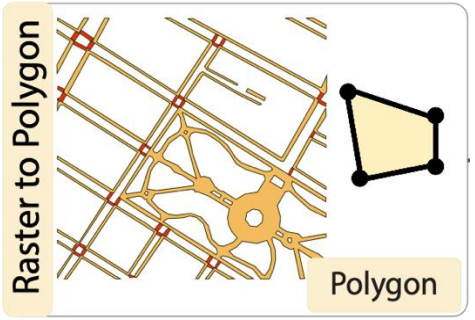


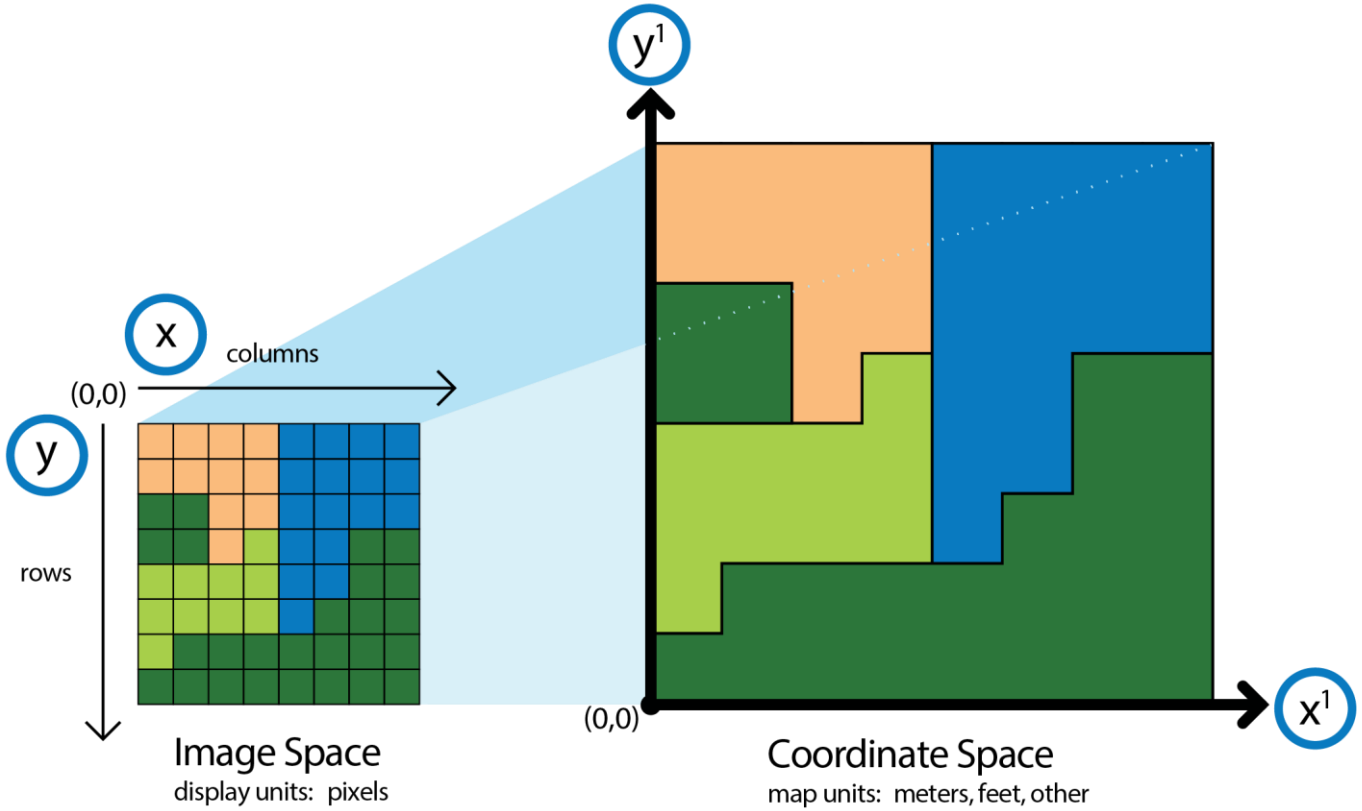


Affine transformation

Image Source: https://www.esri.com/about/newsroom/arcuser/understanding-raster-georeferencing/

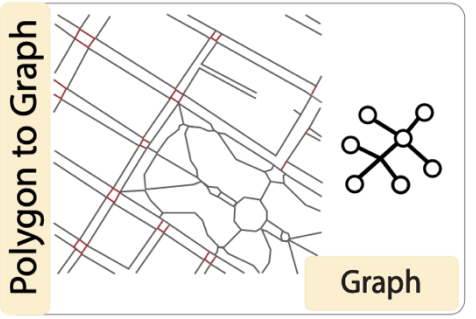
Raster to Geo-referenced Polygon

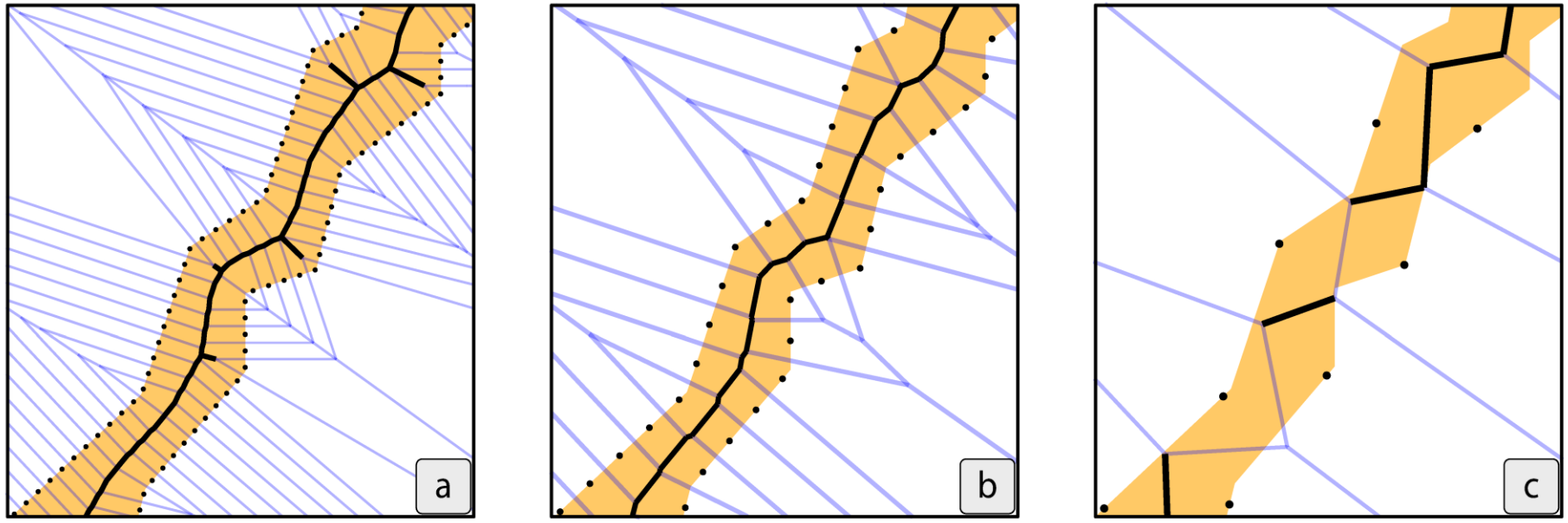




Connected-Component Labeling

Polygon to Line Transformation

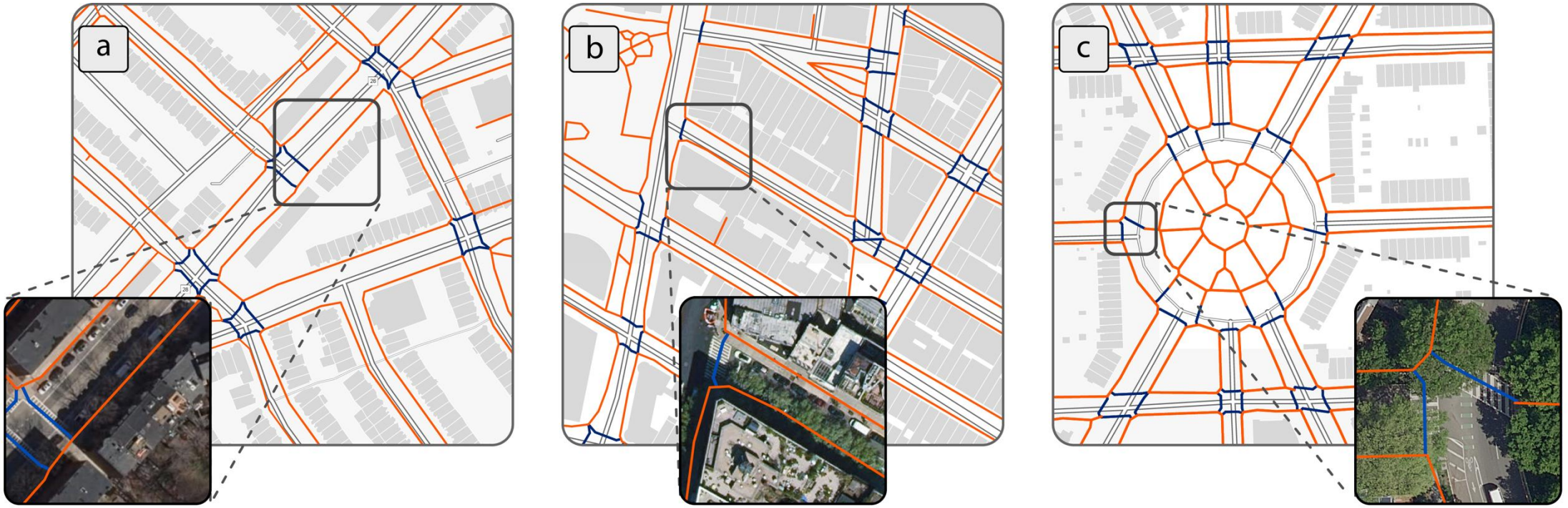




Dense Voronoi method (DV) to create centerline - different interpolation distances **d**

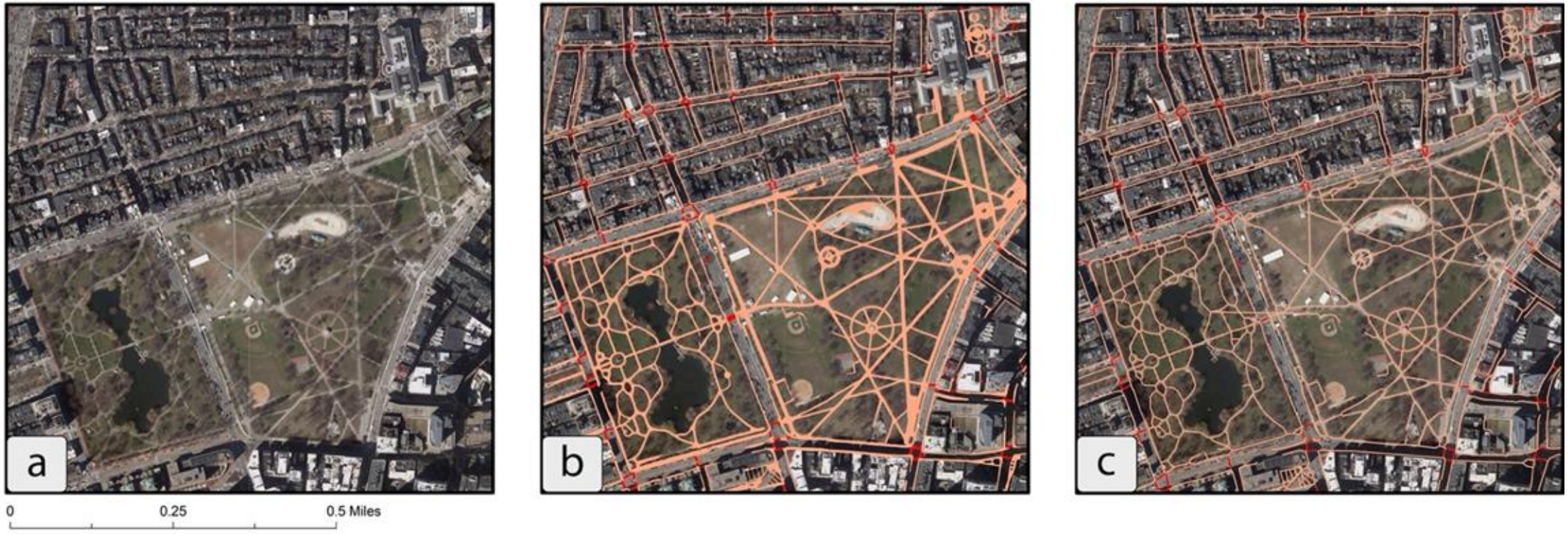
a) d= 0.2 resulting in a smoother line but with more extra branches
b) d= 0.5 created less smooth line but no extra branches,
c) d= 2 resulted in a broken line

Results: Overcoming Obstructions



a) Cambridge, MA. - sidewalks are mapped as continuous despite the heavy shadow
b) Manhattan - sidewalks and crosswalks obstructed by tree foliage and shadow are mapped
c) Washington, DC. - crosswalks covered by vegetation are correctly detected and mapped

Results: Boston Commons



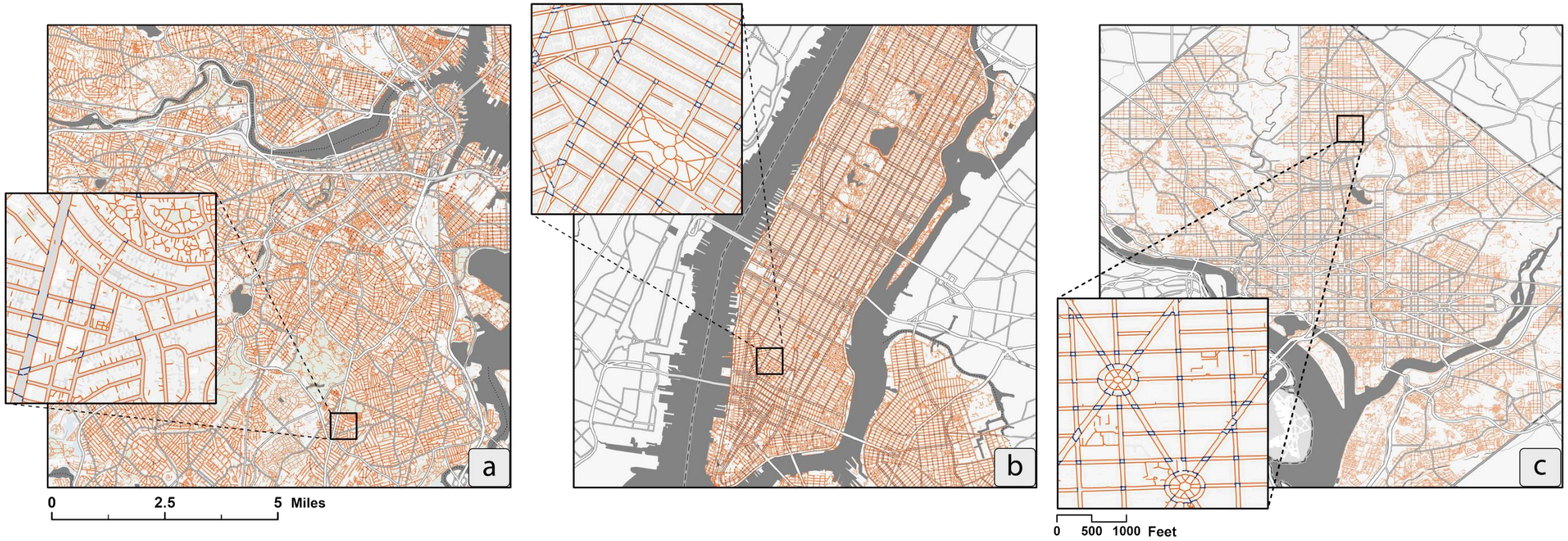
a) Orthorectified aerial tiles stitched together
b) Detected sidewalk and footpath polygons (in orange) and detected crosswalks (in red)
c) Fitted sidewalk, crosswalk, and footpath centerlines superimposed on the aerial image

City-scale Sidewalk Networks

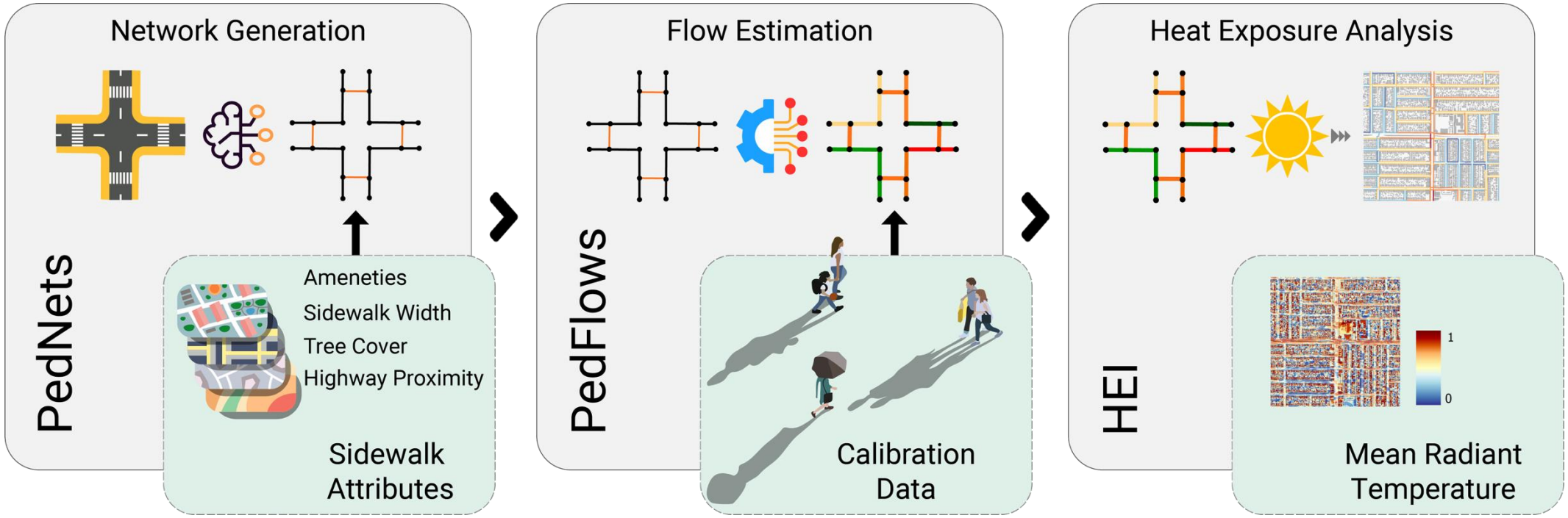
Boston

Manhattan

Washington DC



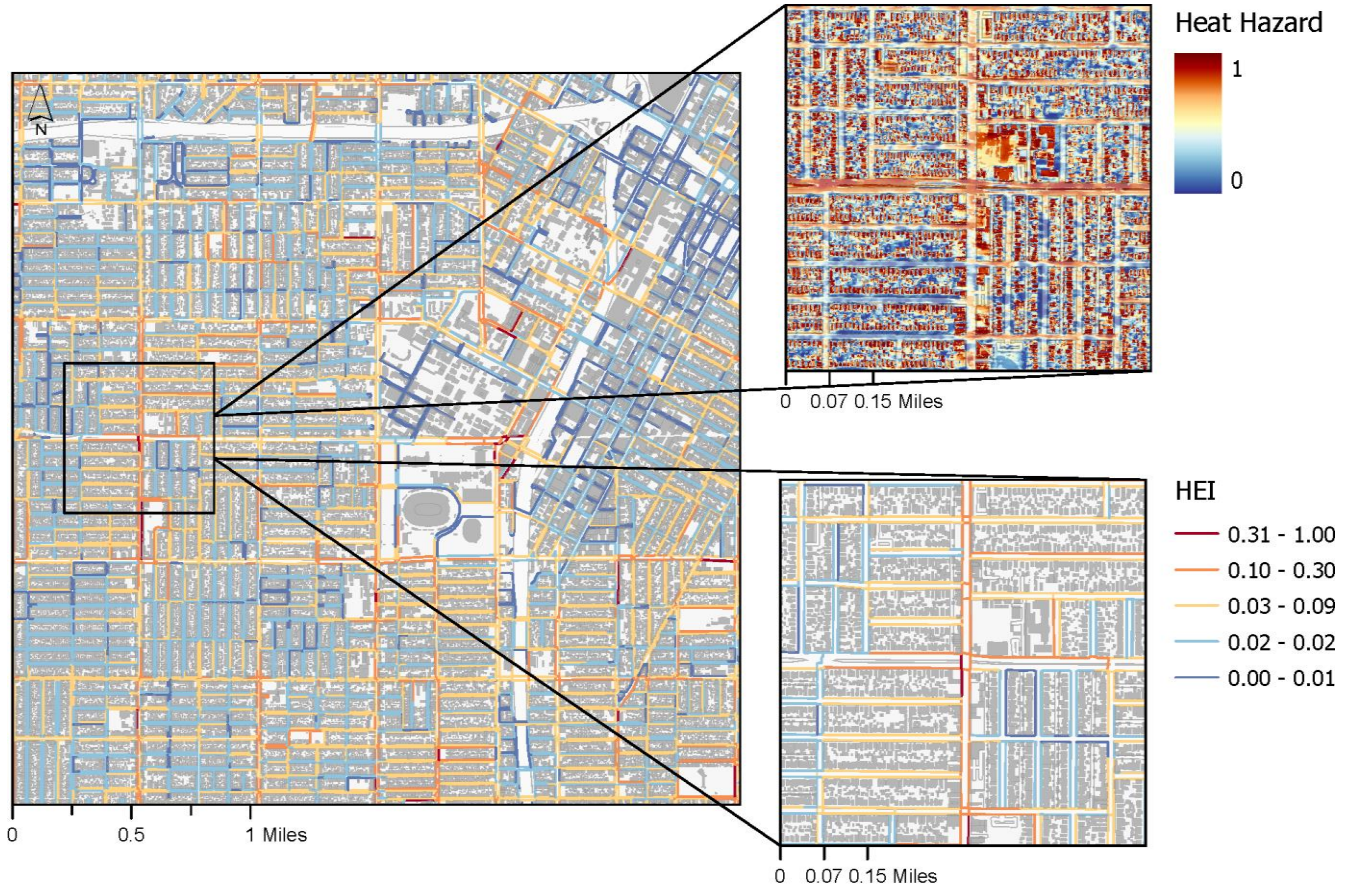
Application: Heat Exposure Index (HEI)



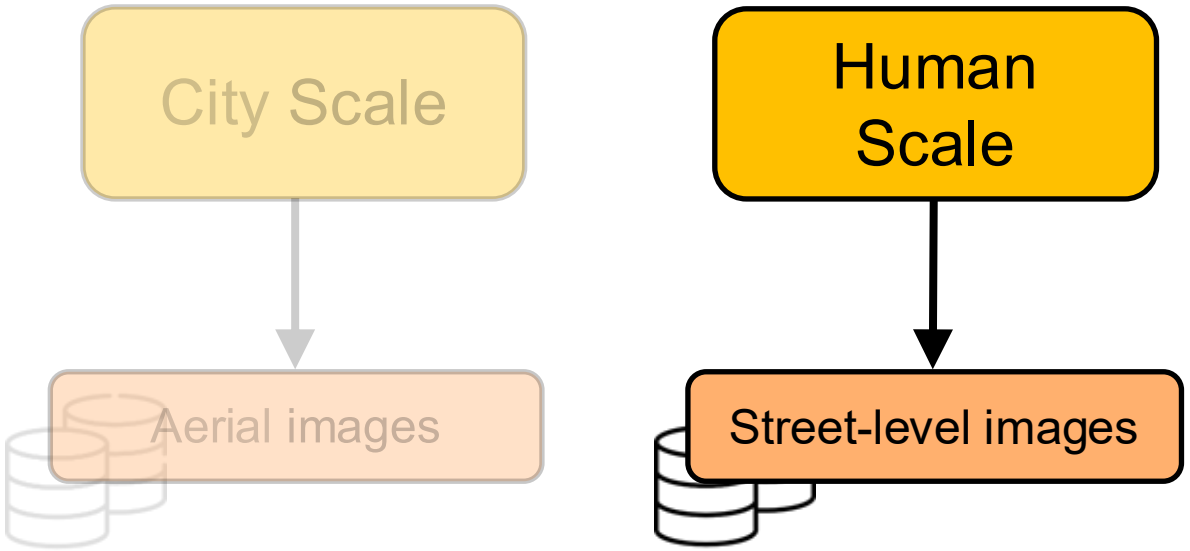
Colaninno, N., R. Basu, M. Hosseini, A. Alhassan, L. Liu, and A. Sevtsuk.
"Urban Heat Risk Assessment: Exploring a Novel Pedestrian Network-Based Framework." In *TRB Annual Meeting*. 2024.

Application: Heat Exposure Index (HEI)

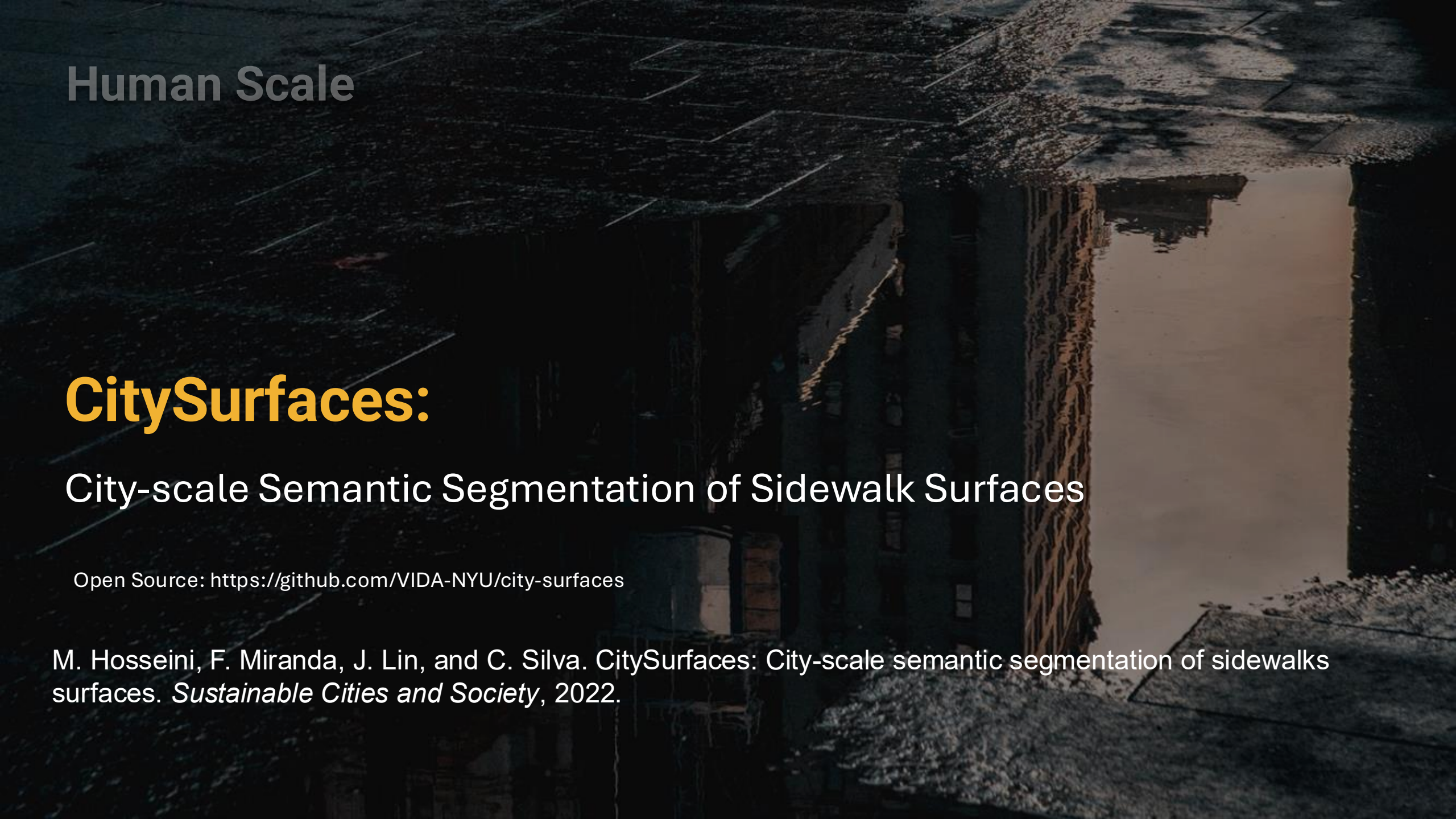
A Heat Exposure Index (HEI) combines heat hazard (mean radiant temperature) and exposure (pedestrian volume) to provide a single measure through which planners
can identify hot spots where a large number of pedestrians might experience high heat stress.



Comprehensive Analysis







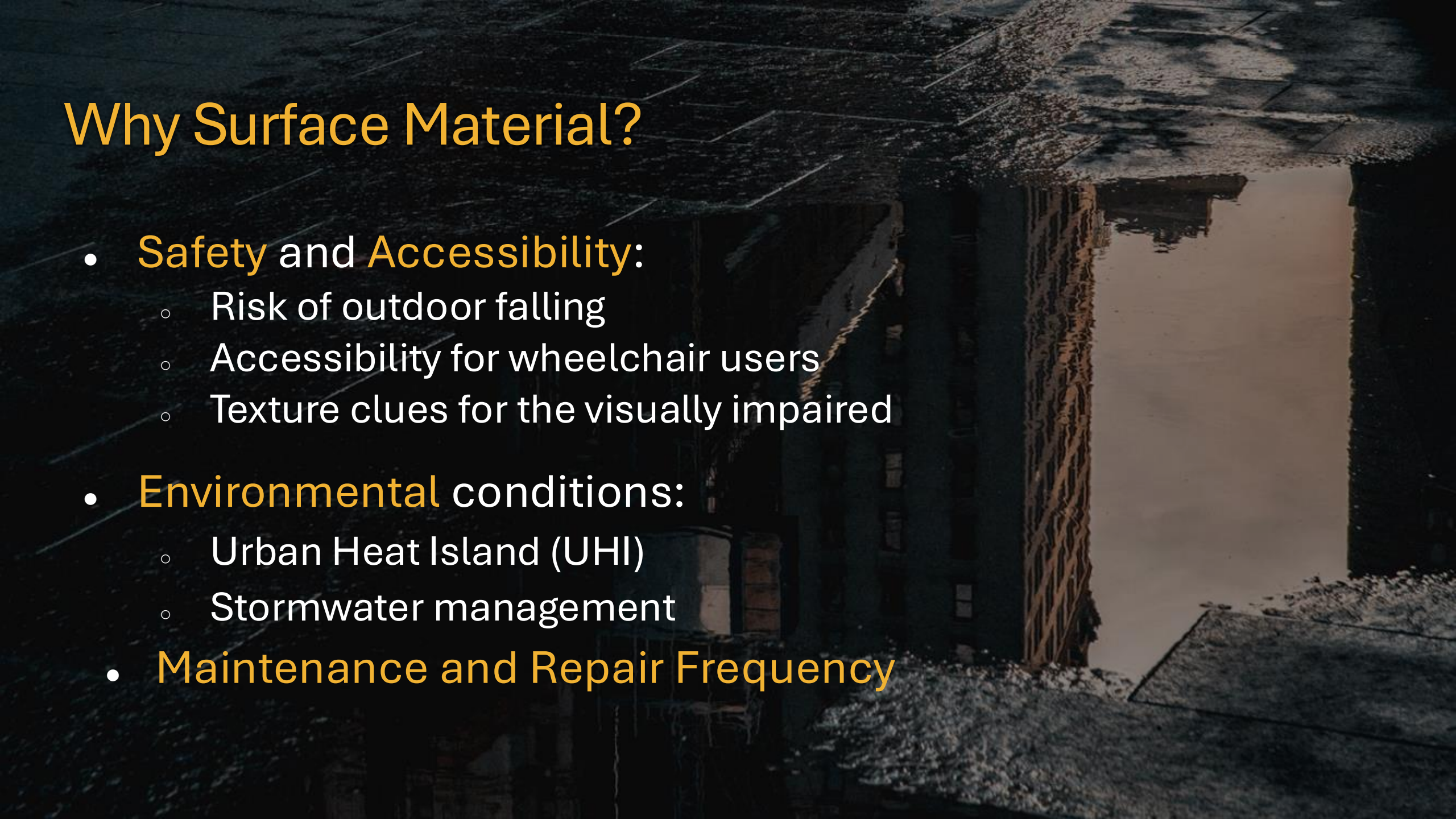
Human Scale

CitySurfaces:

City-scale Semantic Segmentation of Sidewalk Surfaces

Open Source: https://github.com/VIDA-NYU/city-surfaces

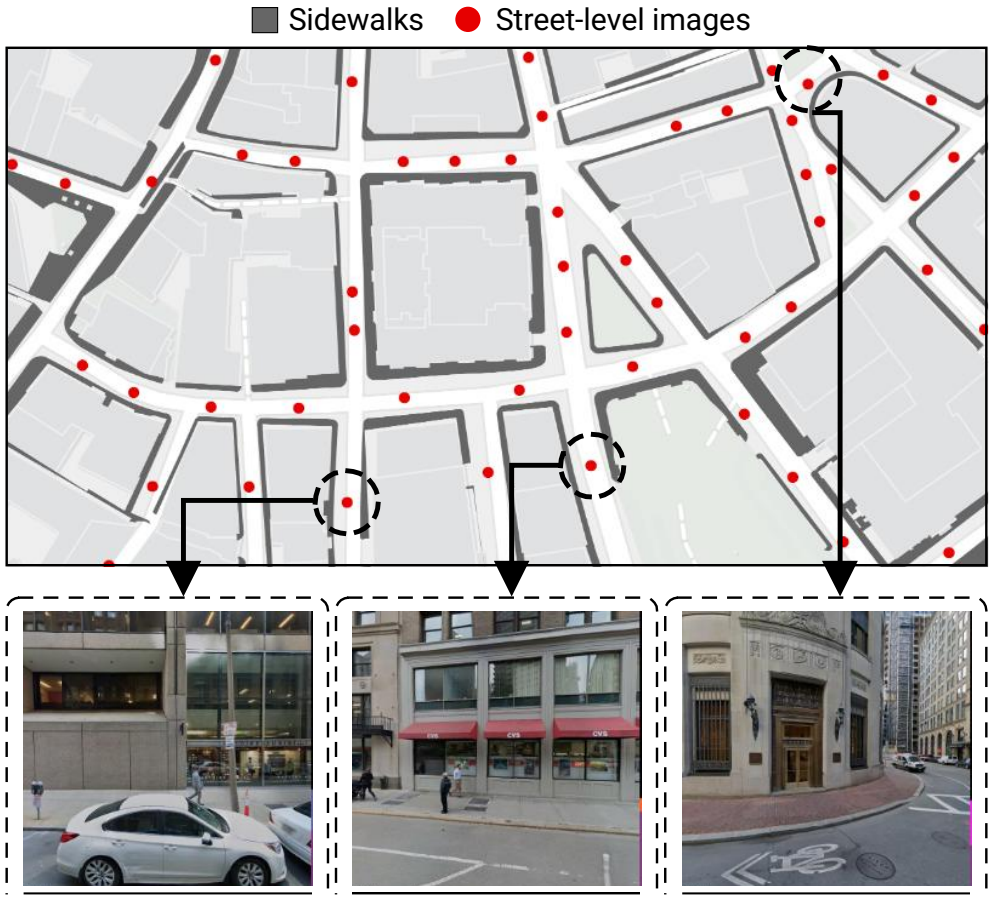
M. Hosseini, F. Miranda, J. Lin, and C. Silva. CitySurfaces: City-scale semantic segmentation of sidewalks surfaces. *Sustainable Cities and Society*, 2022.



Why Surface Material?

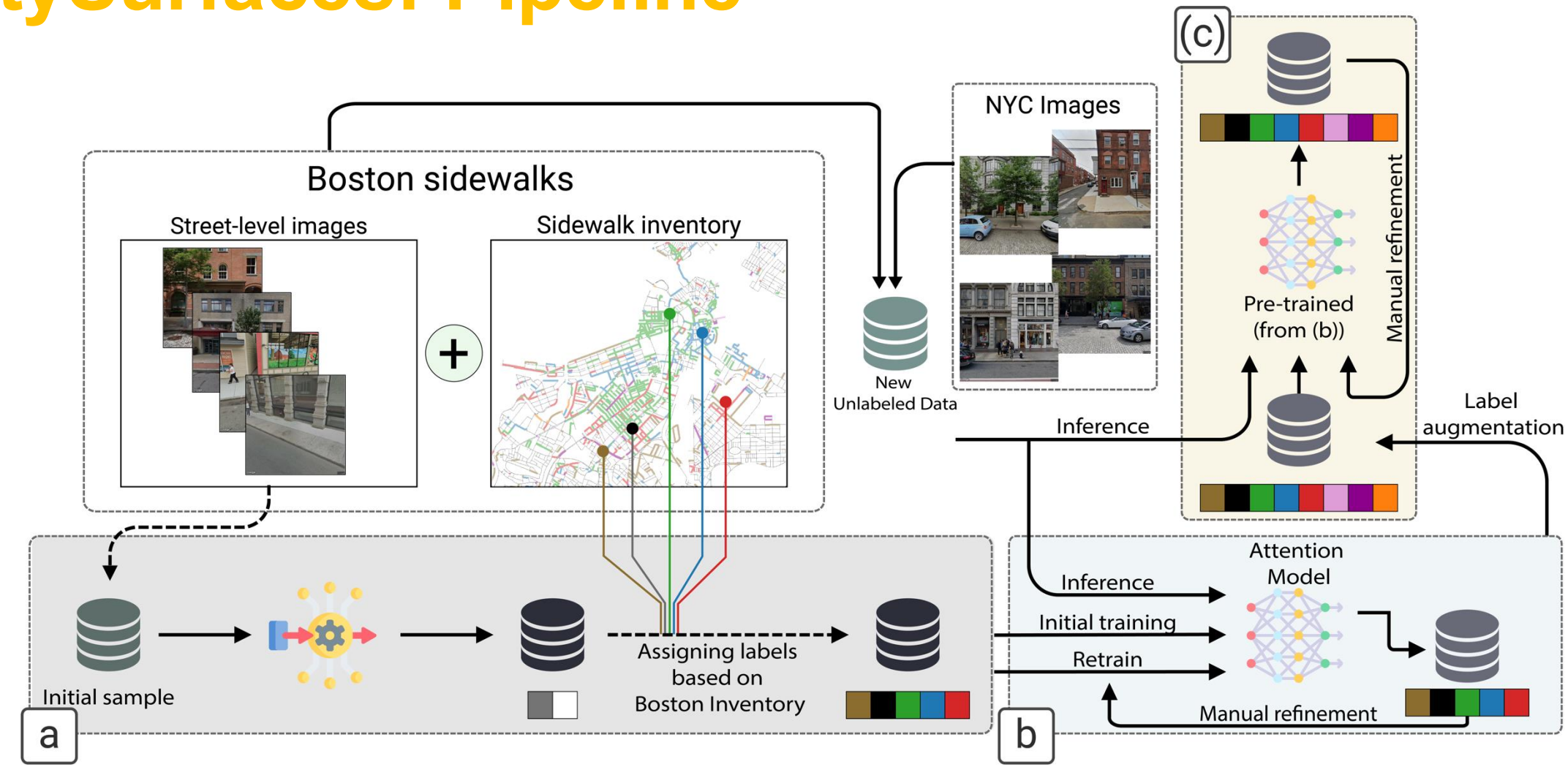
- Safety and Accessibility:
  - Risk of outdoor falling
  - Accessibility for wheelchair users
  - Texture clues for the visually impaired
- Environmental conditions:
  - Urban Heat Island (UHI)
  - Stormwater management
- Maintenance and Repair Frequency

CitySurfaces: Material Classes and Sampling





CitySurfaces: Pipeline



CitySurfaces: Evolution with Active Learning



Input Image

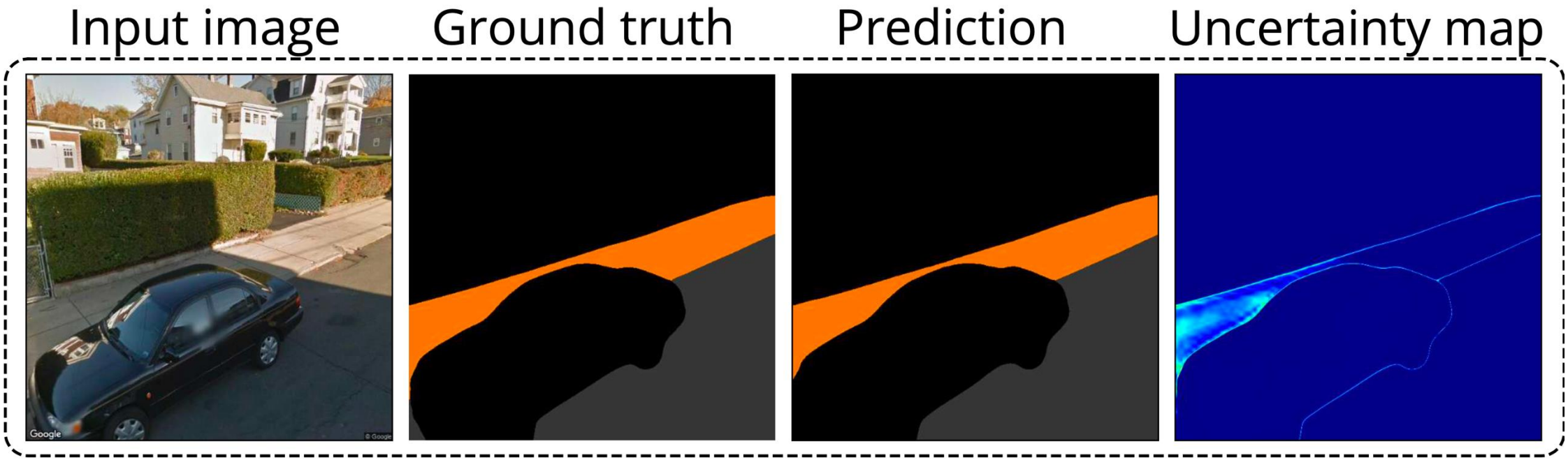


Ground truth



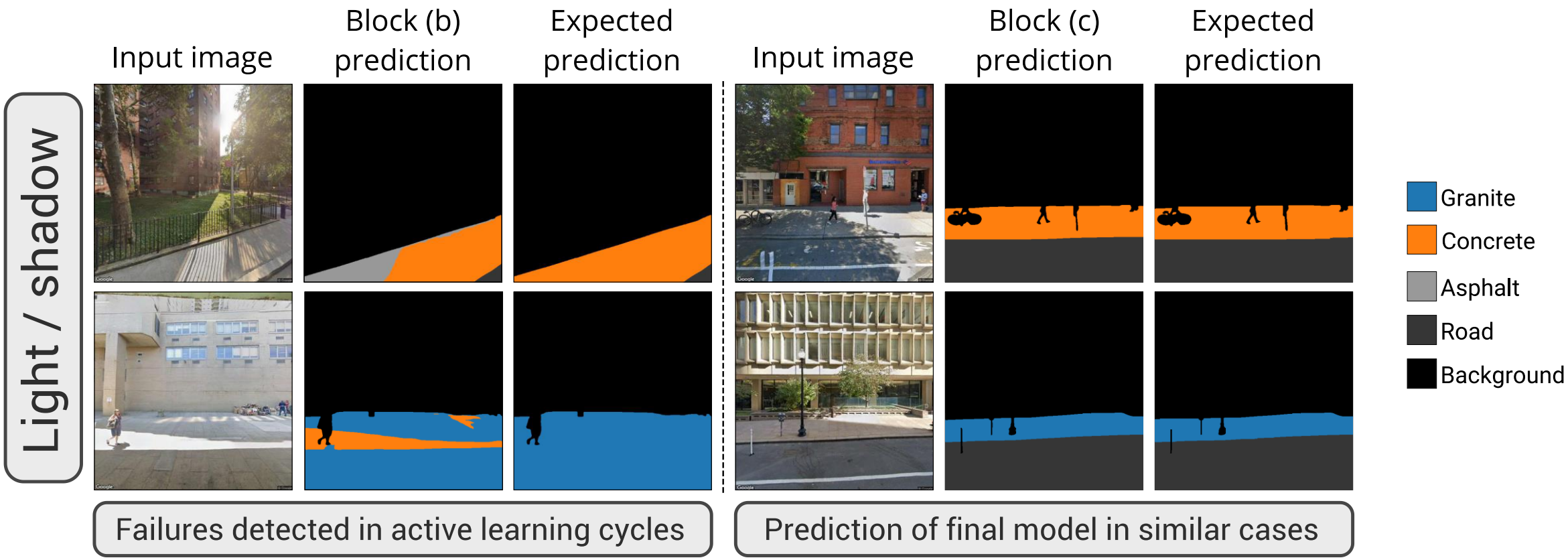
Stage 1 prediction

CitySurfaces: Inside the Blackbox

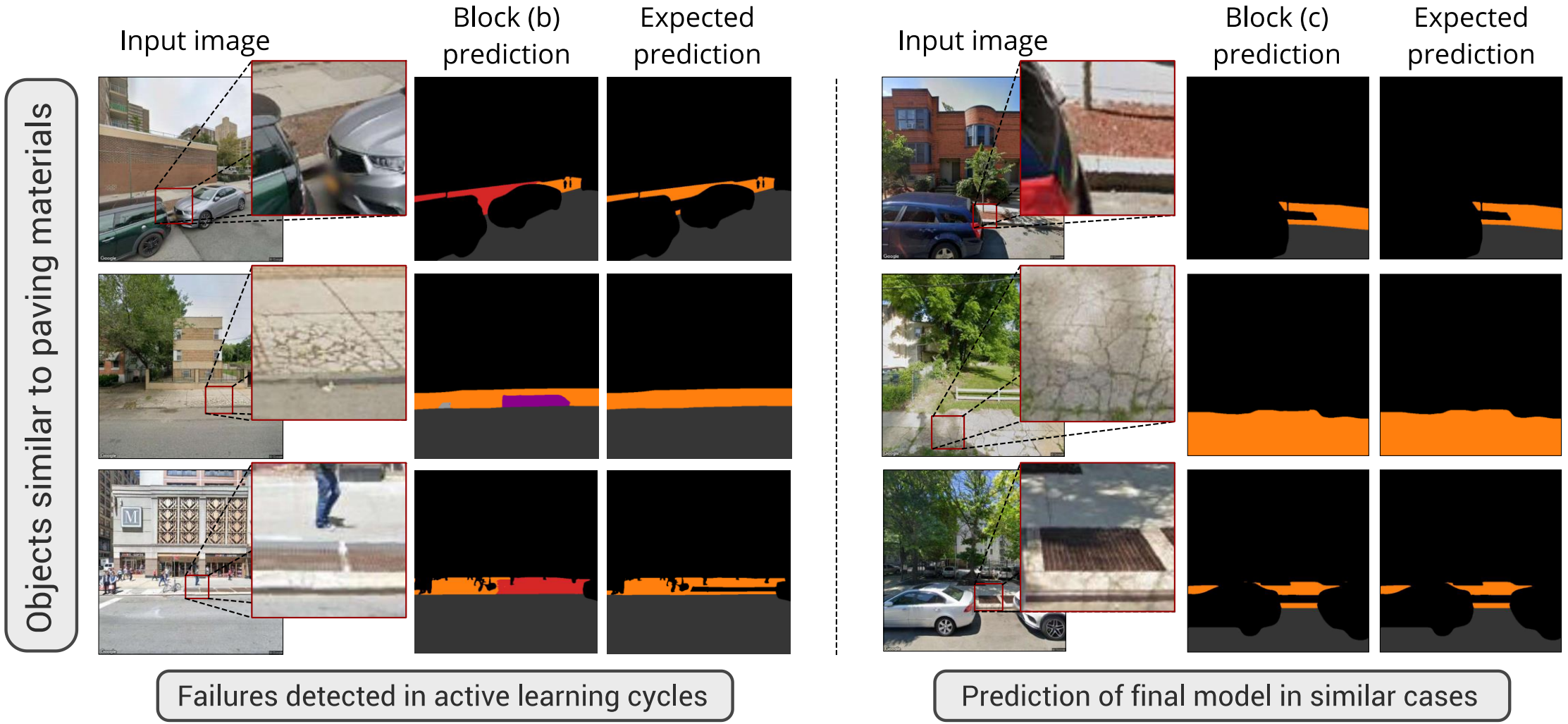


The model's prediction is correct, but the uncertainty map reveals that the model was less certain about the part which is in shadow! So, we need more images like this in our training data

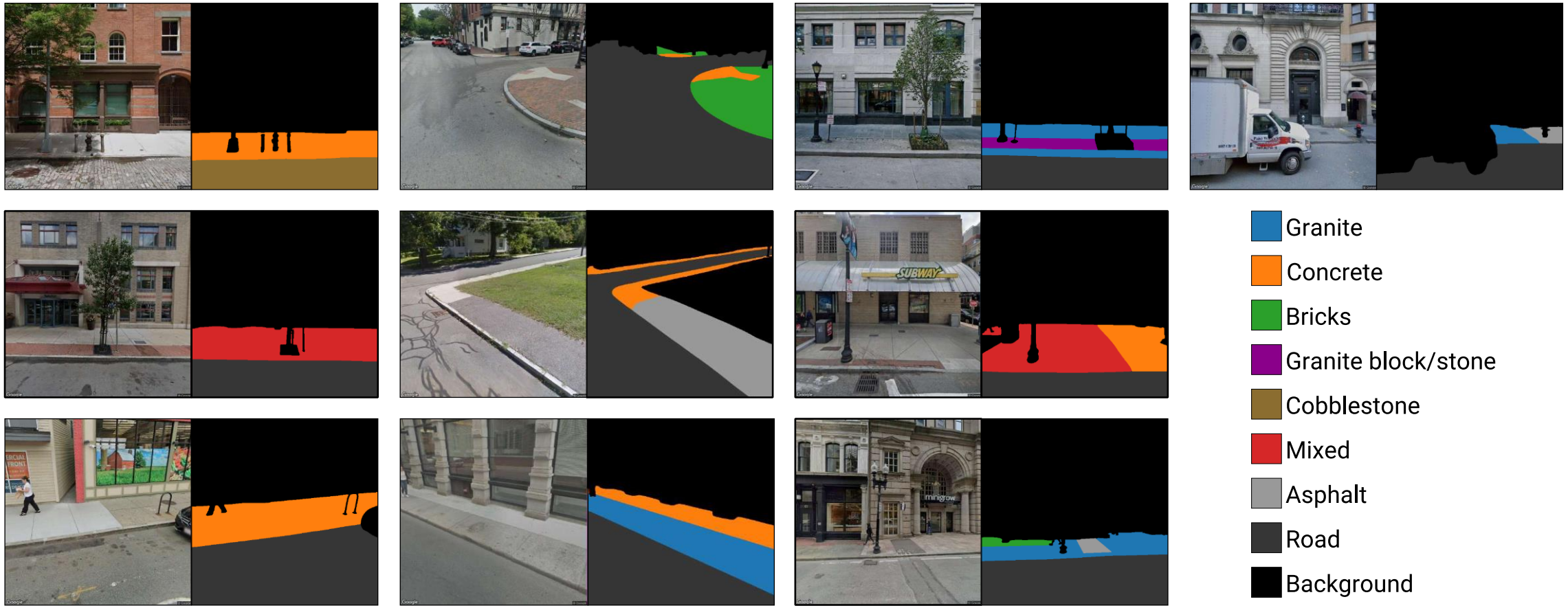
CitySurfaces: Tracking Failures



CitySurfaces: Tracking Failures



CitySurfaces: Results



Surface Materials Applications



Creating sidewalk inventories for different cities



Accessibility-aware routing apps



Fall prevention programs

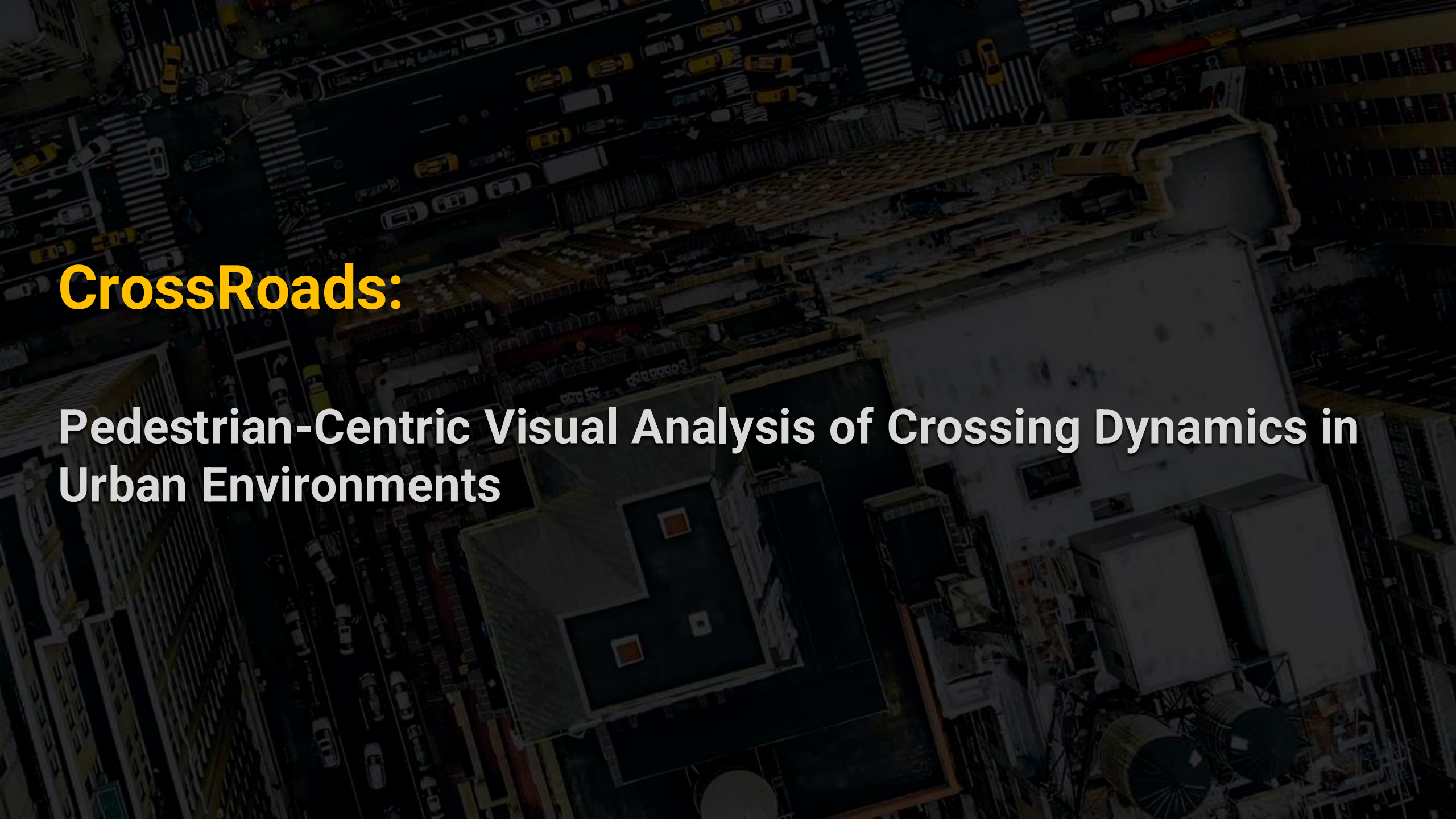


Water Runoff Management and Urban Heat Island effect

Data Collection

- Data collection impacted by:
  - Who is collecting the data?
  - Why this data is being collected?
  - How the data is being collected?
- We need open-source, Findable, Accessible, Interoperable, Reusable (FAIR) data
- As we are innovating, we are also setting norms and we should be conscious and diligent about the norms we are setting

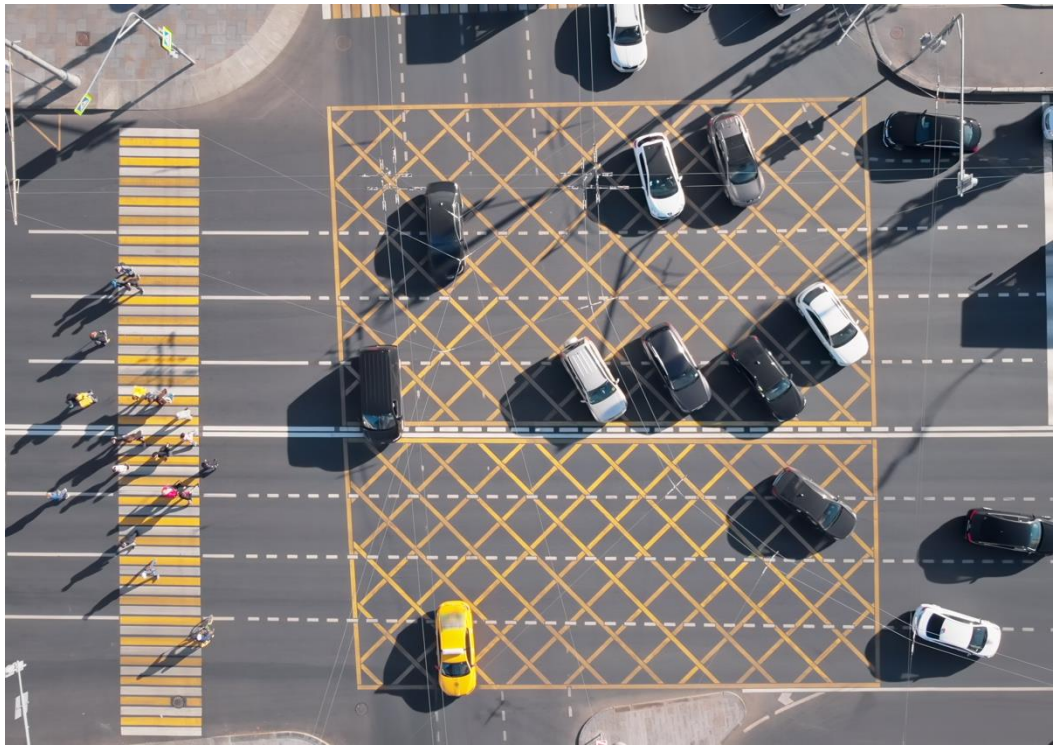




CrossRoads:

Pedestrian-Centric Visual Analysis of Crossing Dynamics in Urban Environments

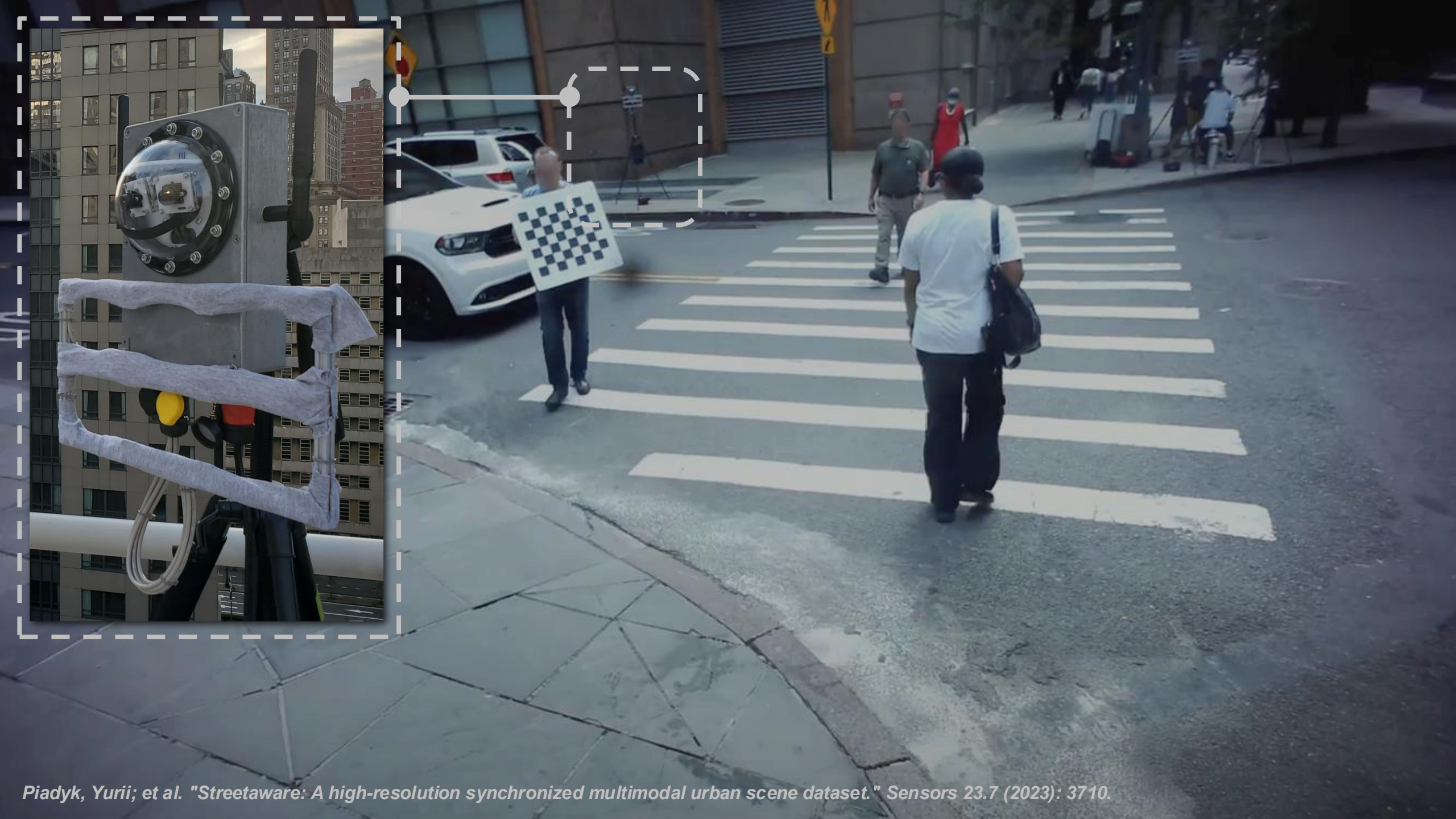
Top-down vs. Pedestrian View



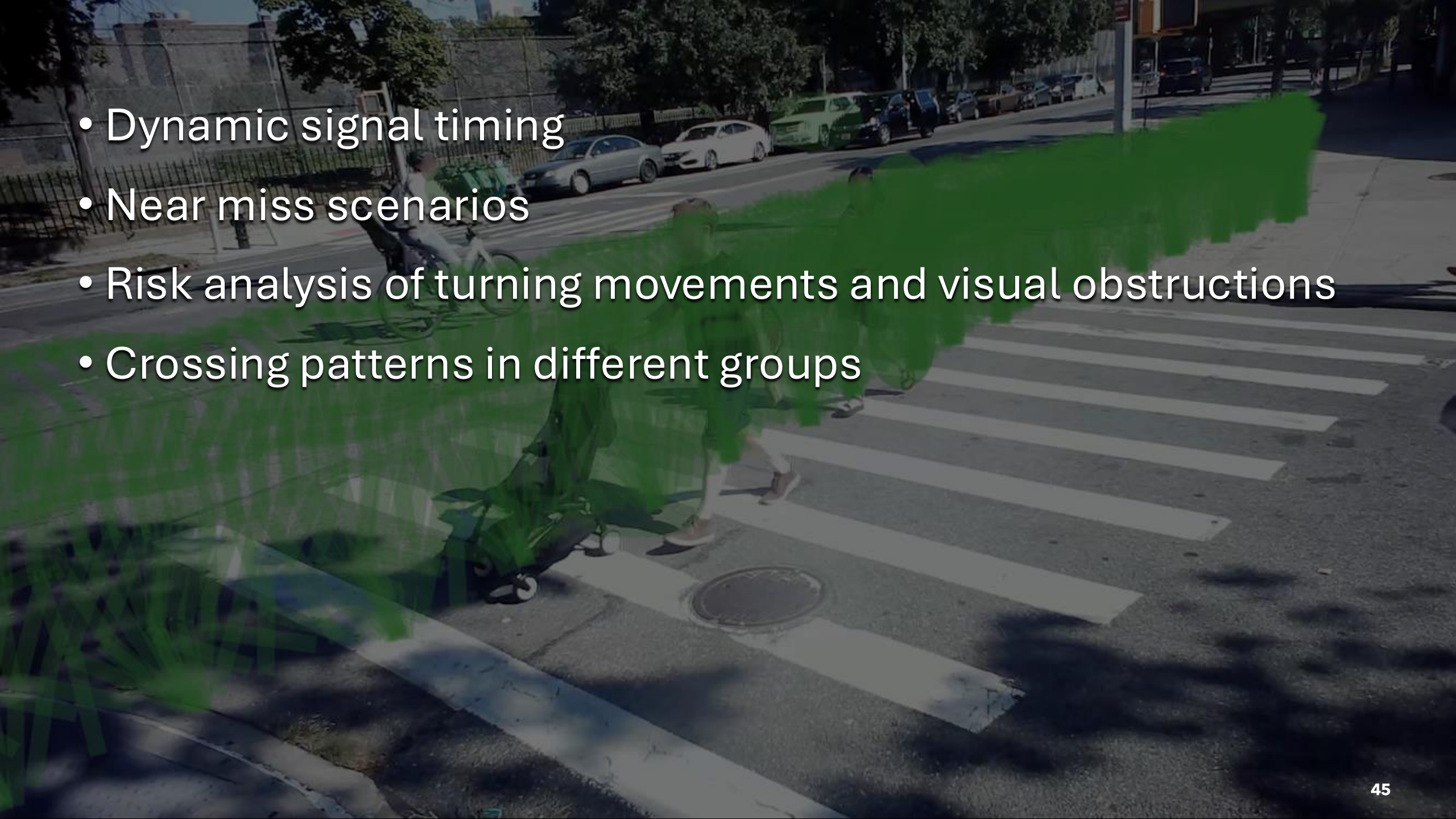
Top-down



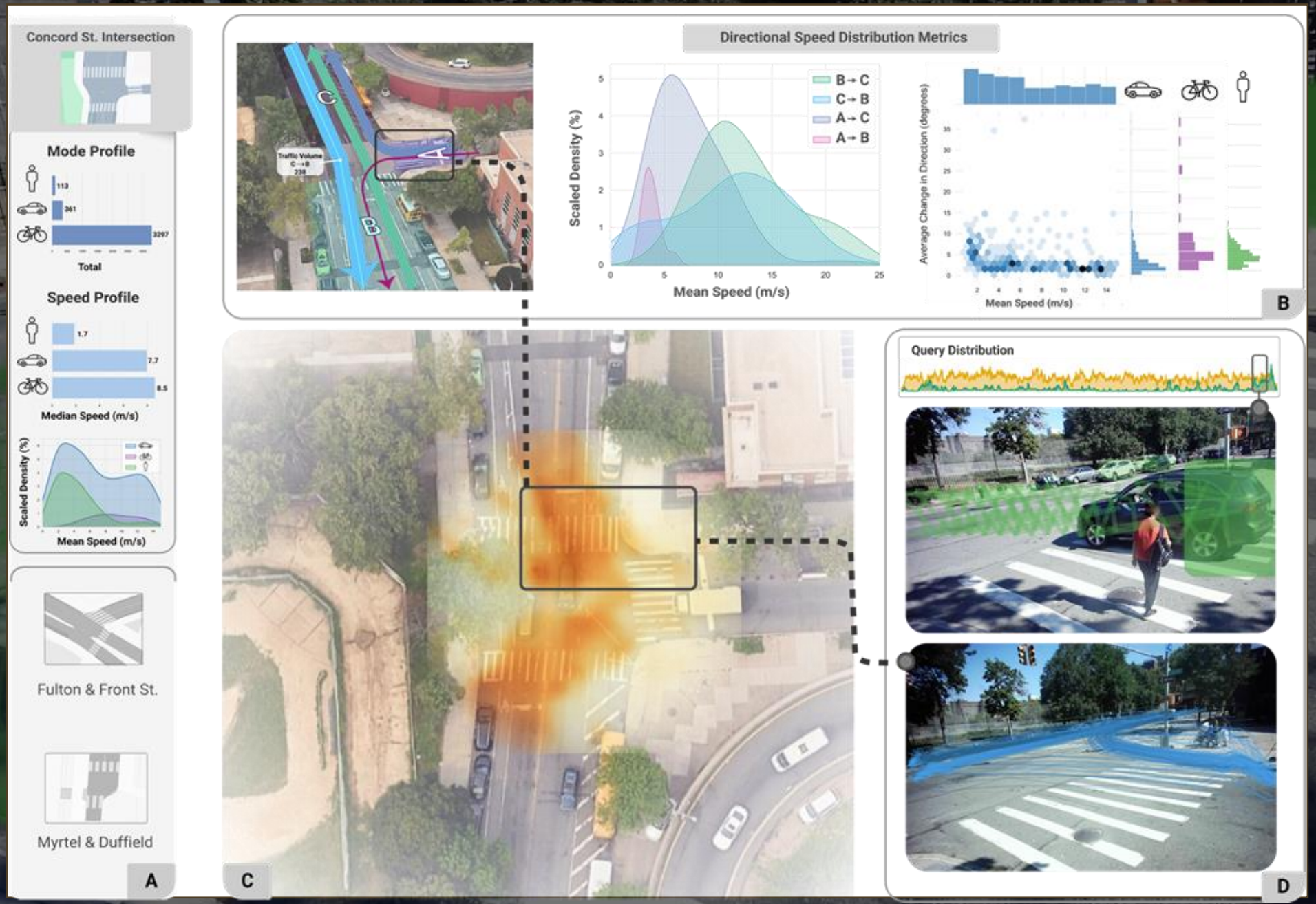
Pedestrian View



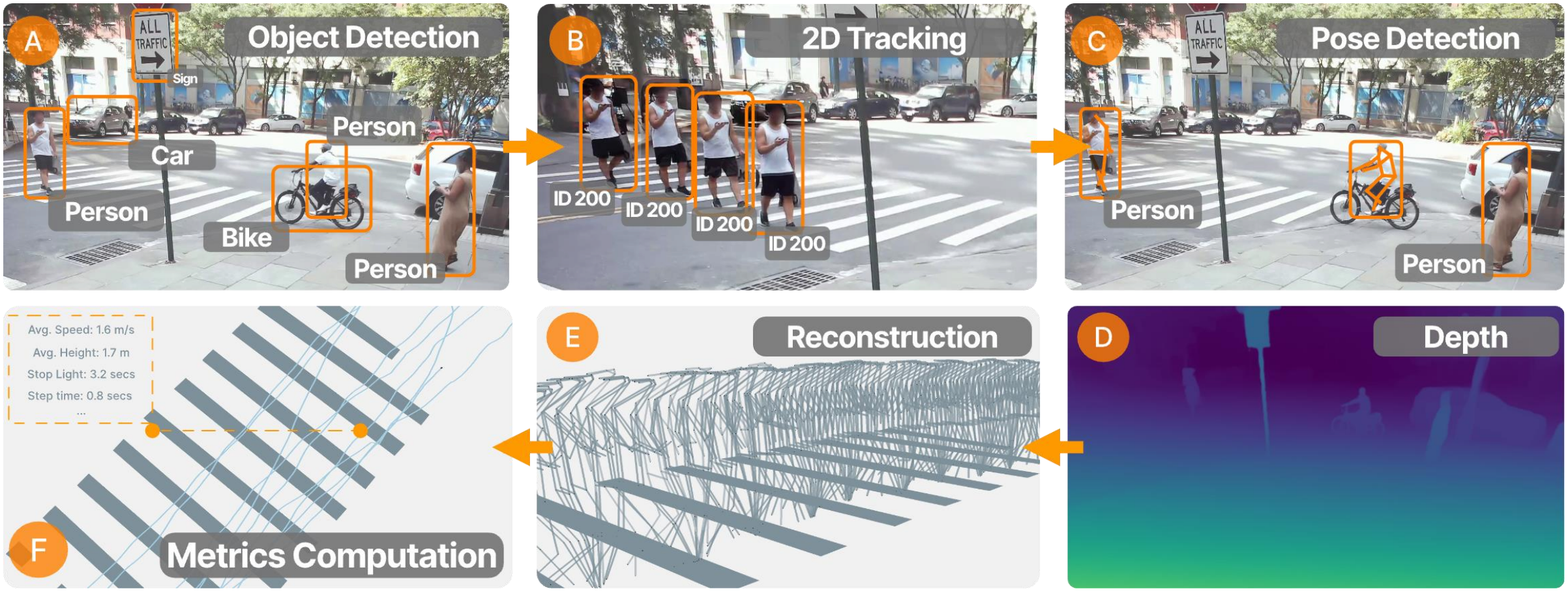
Piadyk, Yurii; et al. "Streetaware: A high-resolution synchronized multimodal urban scene dataset." Sensors 23.7 (2023): 3710.



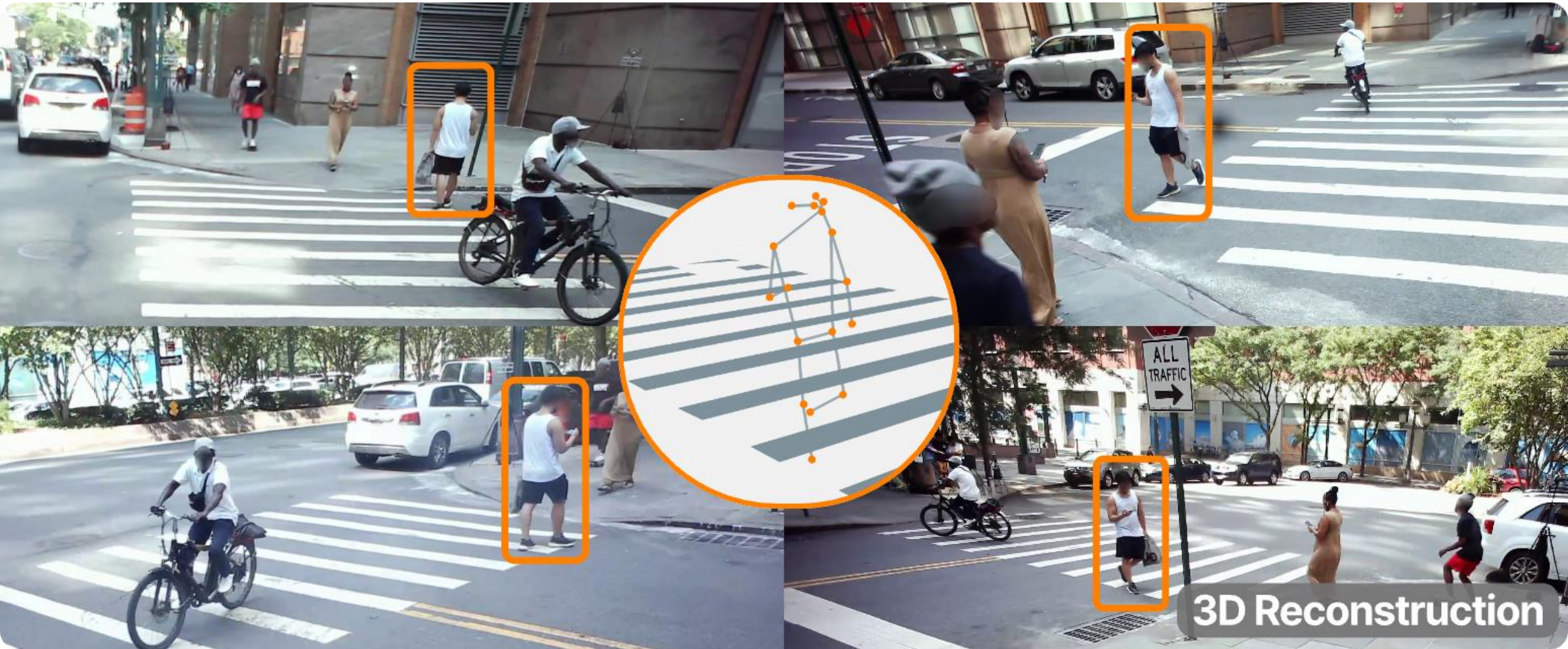
- Dynamic signal timing
- Near miss scenarios
- Risk analysis of turning movements and visual obstructions
- Crossing patterns in different groups


Concord St. Intersection
Mode Profile
113
361
3297
Total
Speed Profile
1.7
7.7
8.5
Median Speed (m/s)
Scaled Density (%)
Mean Speed (m/s)
Fulton & Front St.
Myrtel & Duffield
A
C
B
Traffic Volume
C → B
238
Directional Speed Distribution Metrics
B → C
C → B
A → C
A → B
Scaled Density (%)
Mean Speed (m/s)
Average Change in Direction (degrees)
Mean Speed (m/s)
B
Query Distribution
D

CrossRoad: Analysis Pipeline



CrossRoad: 3D Reconstruction






Baby Carrier
Stroller
Mobility Scooter
Walking Cane

Factors Influencing Walking Speed



Natural Pace



Footwear



Luggages



Terrain



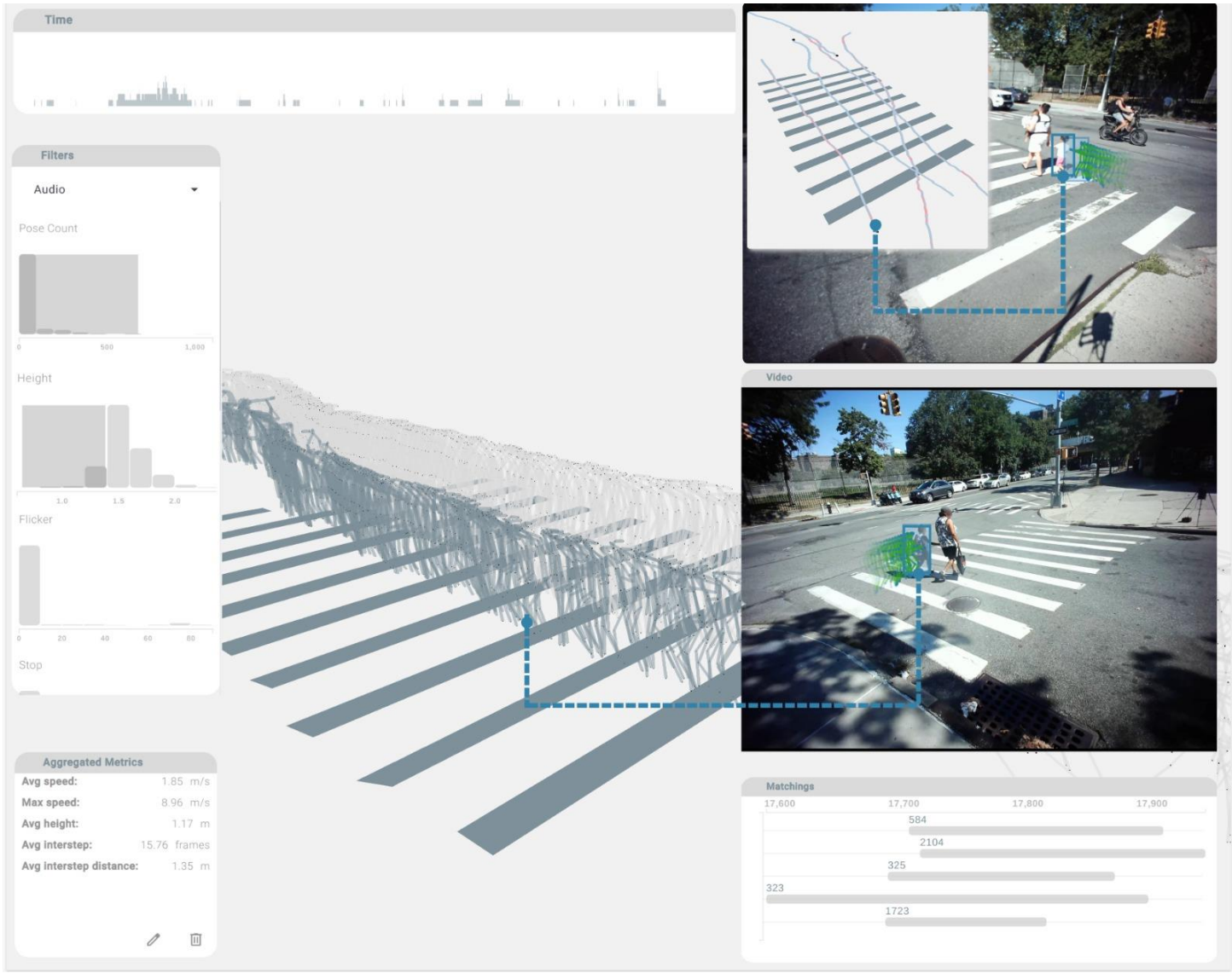
Pedestrian Flow

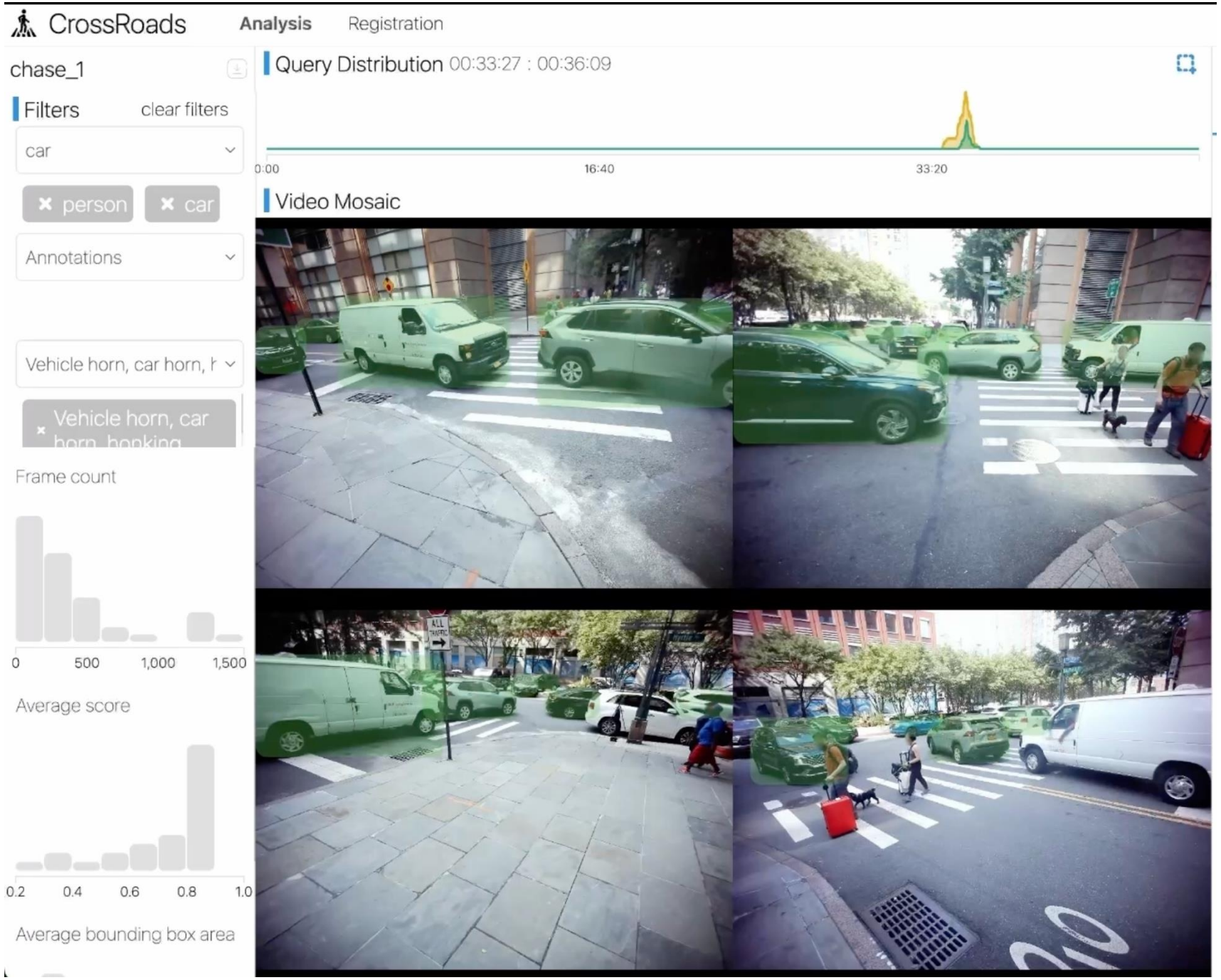


Route Familiarity






Time
Filters
Audio
Pose Count
0
500
1,000
Height
1.0
1.5
2.0
Flicker
0
20
40
60
80
Stop
Aggregated Metrics
Avg speed: 1.85 m/s
Max speed: 8.96 m/s
Avg height: 1.17 m
Avg interstep: 15.76 frames
Avg interstep distance: 1.35 m
Video
Matchings
17,600
17,700
17,800
17,900
584
2104
325
323
1723

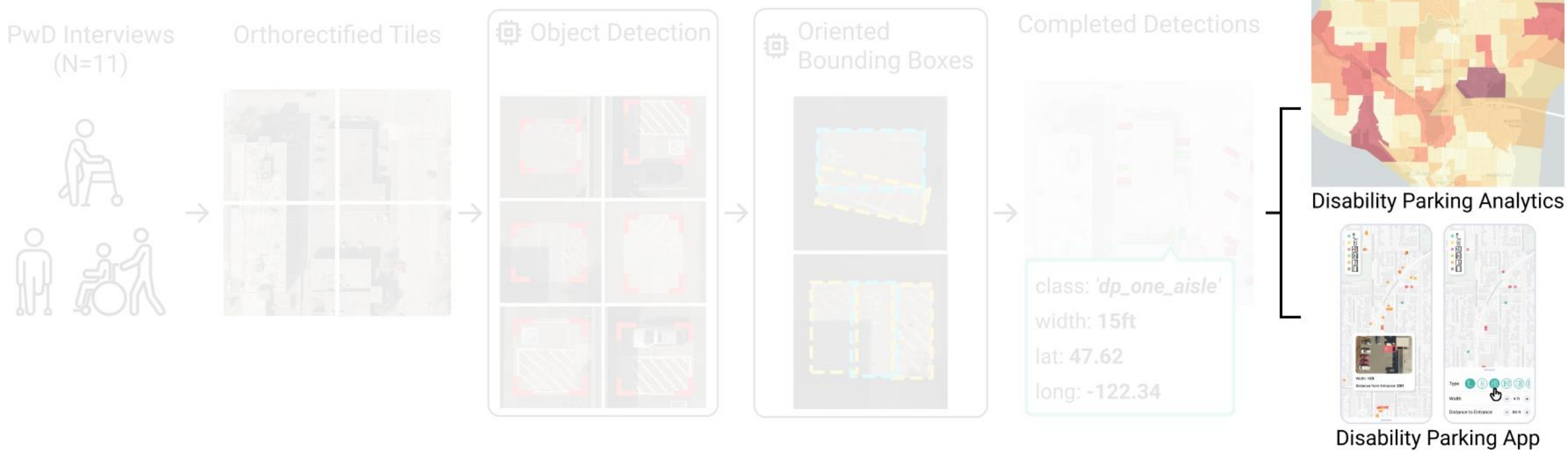

CrossRoads
Analysis
Registration
chase_1
Filters
clear filters
car
person
car
Annotations
Vehicle horn, car horn, h
Vehicle horn, car horn, honking
Frame count
0
500
1,000
1,500
Average score
0.2
0.4
0.6
0.8
1.0
Average bounding box area
Query Distribution 00:33:27 : 00:36:09
0:00
16:40
33:20
Video Mosaic


A
Pose Detection
B
2D Tracking
ID 01
ID 12
2D
C
Camera Registration
SIFT
Metashape
D
3D Reconstruction
Target
1st result
2nd result
3rd result
4th result
3D

Needfinding

Algorithmic Pipeline and Dataset

New Applications



Interview Study

How do PwDs use disability parking?

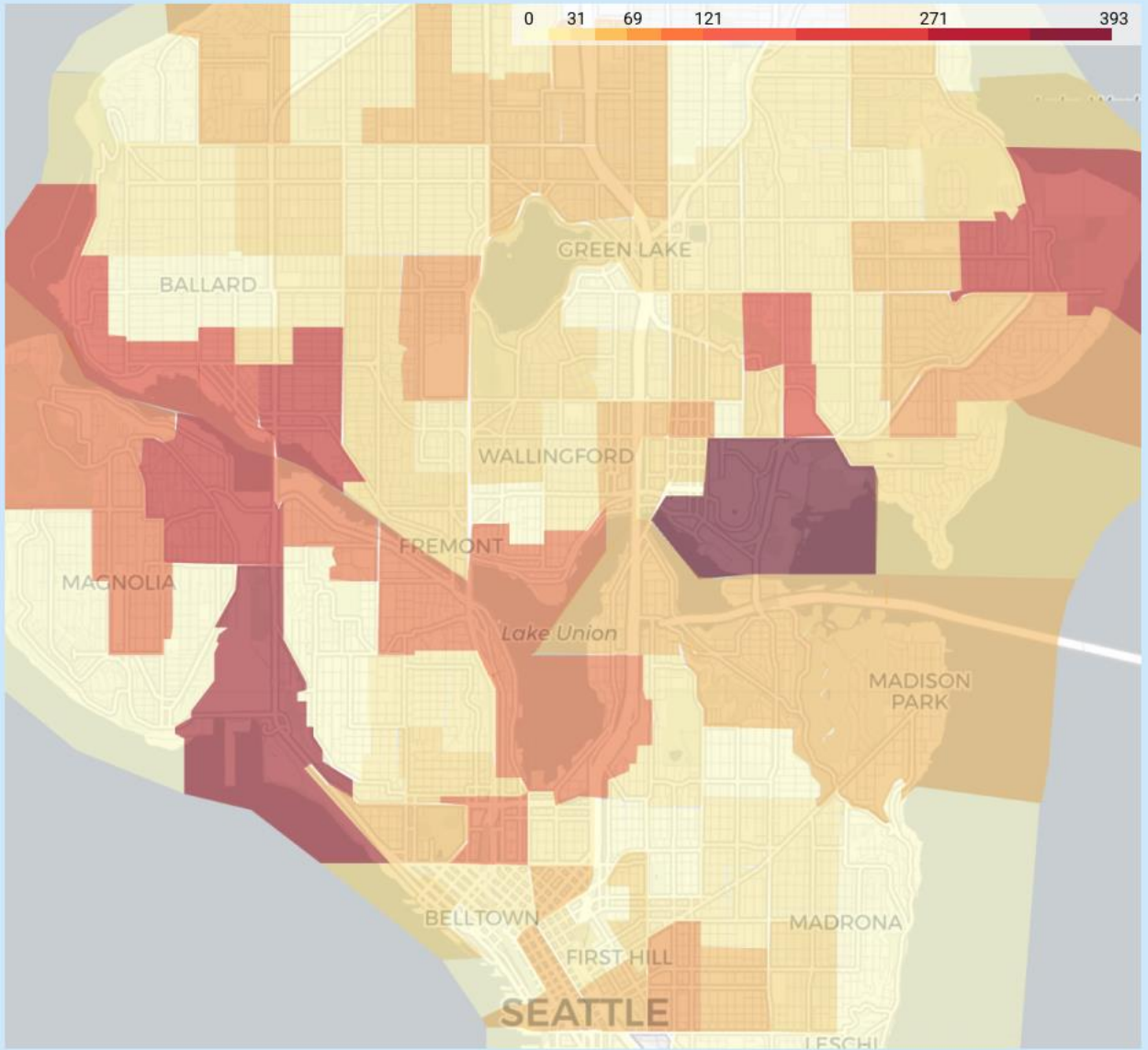
Dataset & Pipeline Development

How well can CV models locate and characterize disability parking from aerial imagery?

New Applications

How could our CV model be used to build useful technology tools?

Policy Level

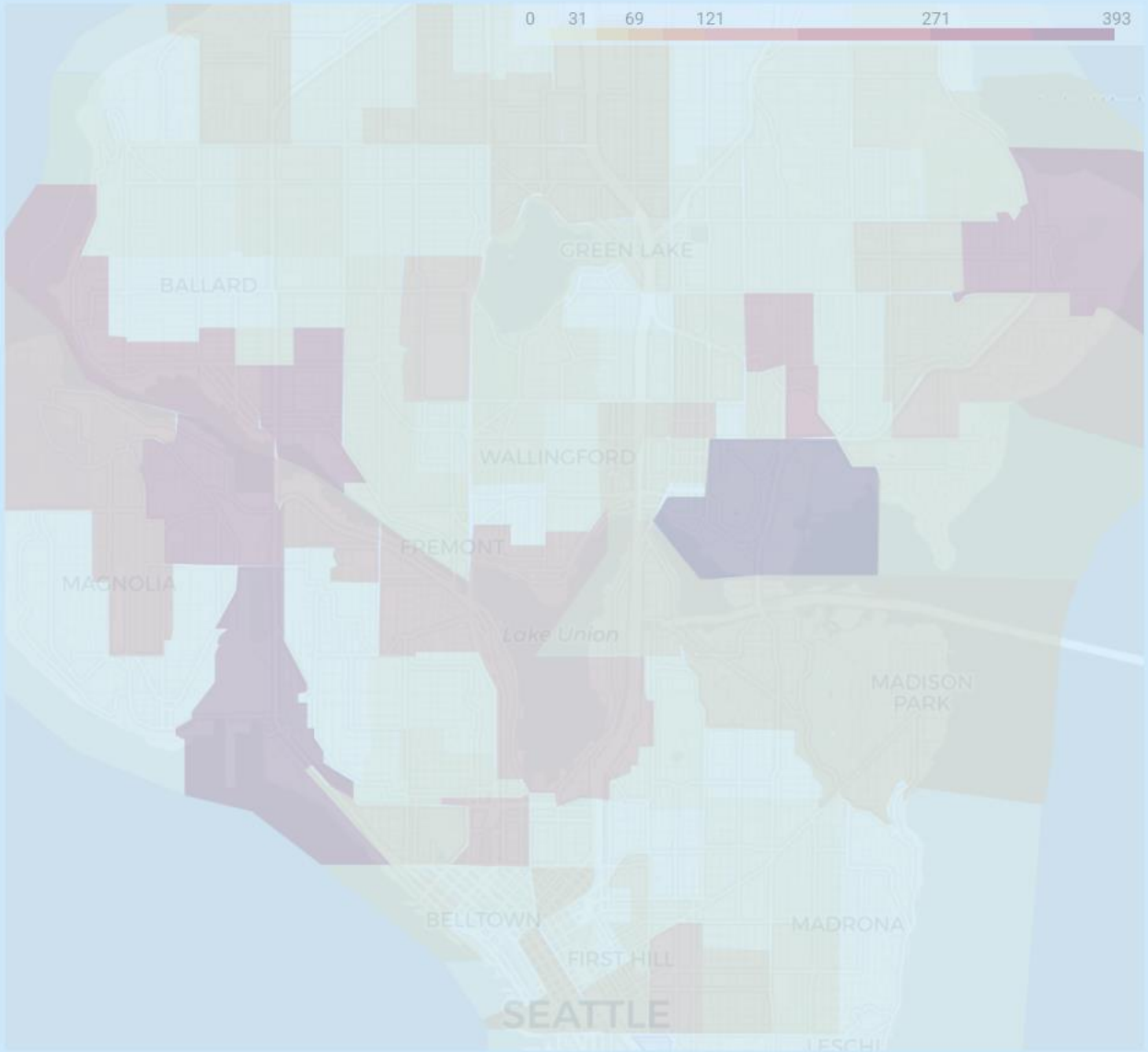


Individual Level





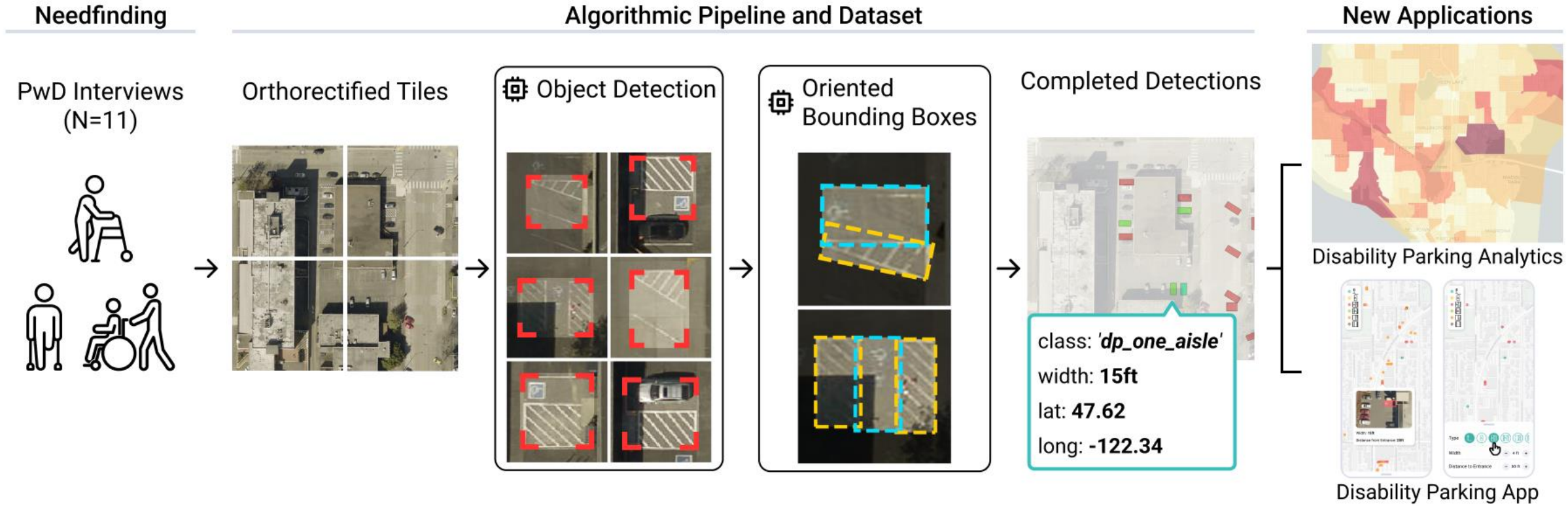
Policy Level



Individual Level







Interview Study

→ Formative findings on how PwDs navigate disability parking, alongside policy recommendations

Dataset & Pipeline Development

→ An open-source dataset of 11,762 labeled disability parking objects

New Applications

→ A CV tool that locates, classifies, and characterizes disability parking from aerial imagery

Questions?

